

EMCH380 – Project Management

Project Scheduling

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Slides are developed from instructor's notes and other references.

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Critical Path Method (CPM)

Project Management – Project Scheduling

PROJECT SCHEDULING

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- Perform Network Calculations:
 - Total Time Needed for Project Completion
 - Critical Path & Critical Activities
 - Activity Total and Free Float

NETWORK TECHNIQUES

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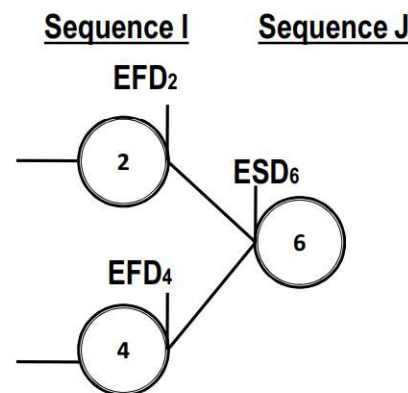
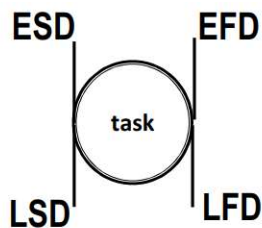
- Critical Path Methods (CPM)
- Program Evaluation and Review Techniques (PERT)

CRITICAL PATH METHODS

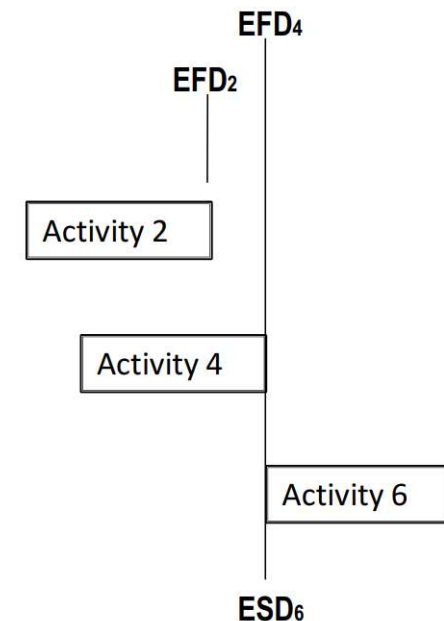
- Designed for construction projects
- Uses deterministic activity time estimates to control both time and cost aspects of a project
- In CPM activities can be crashed (expedited) at extra cost to speed up completion time of project

AON NETWORK COMPUTATIONS – DETERMINATION OF EARLY TIMES

- ESD: Early start date
- EFD: Early finish date
- ESD_0 : Assigned by scheduler,
- $EFD_0 = ESD_0 + T_0$



$$ESD_J = \text{Max } EFD_I \quad \forall I$$



Max implies that the maximization
 $\forall J$ is to be over all links IJ terminating
 at activity J

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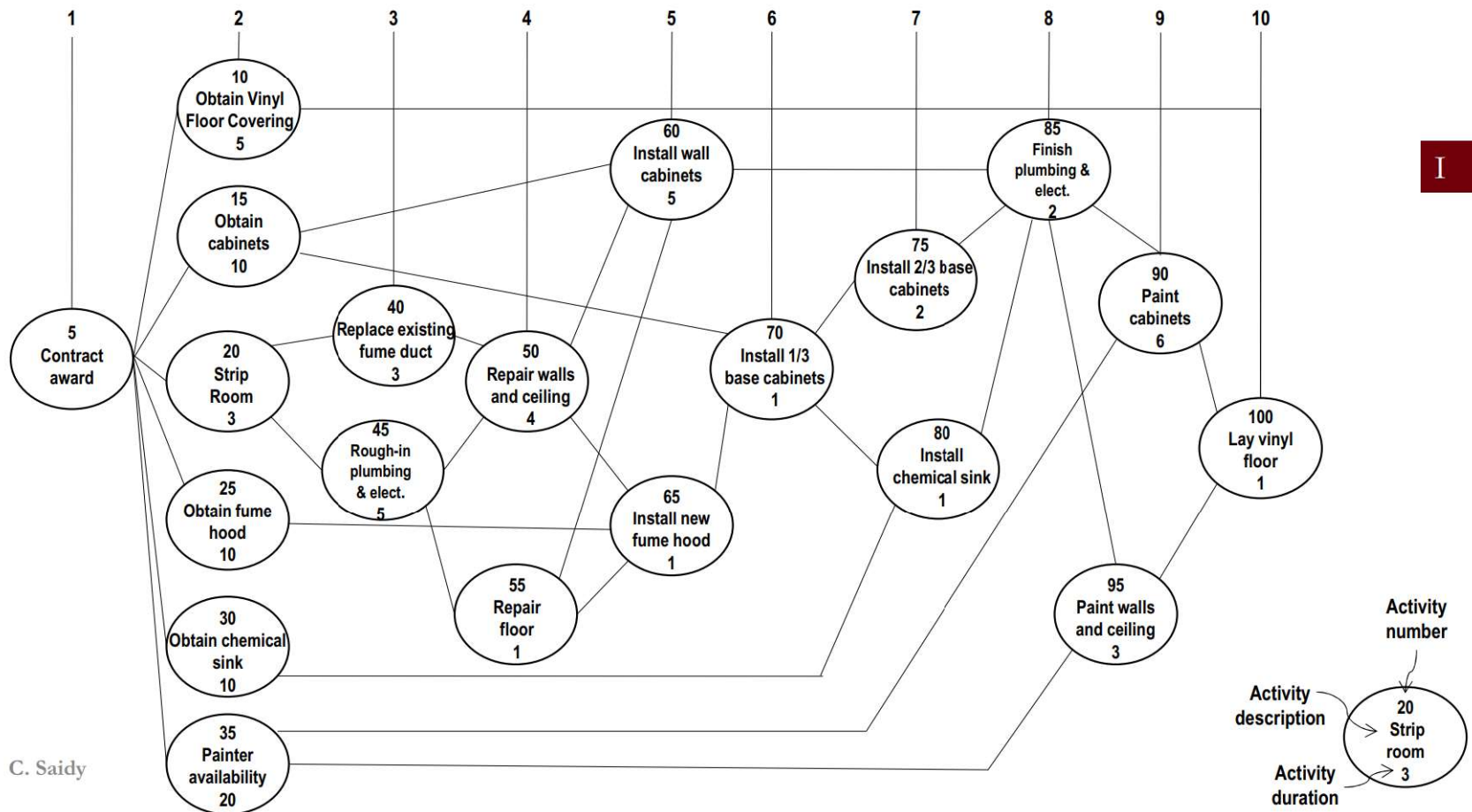
AON – EARLY TIME COMPUTATIONS

► Forward pass computations

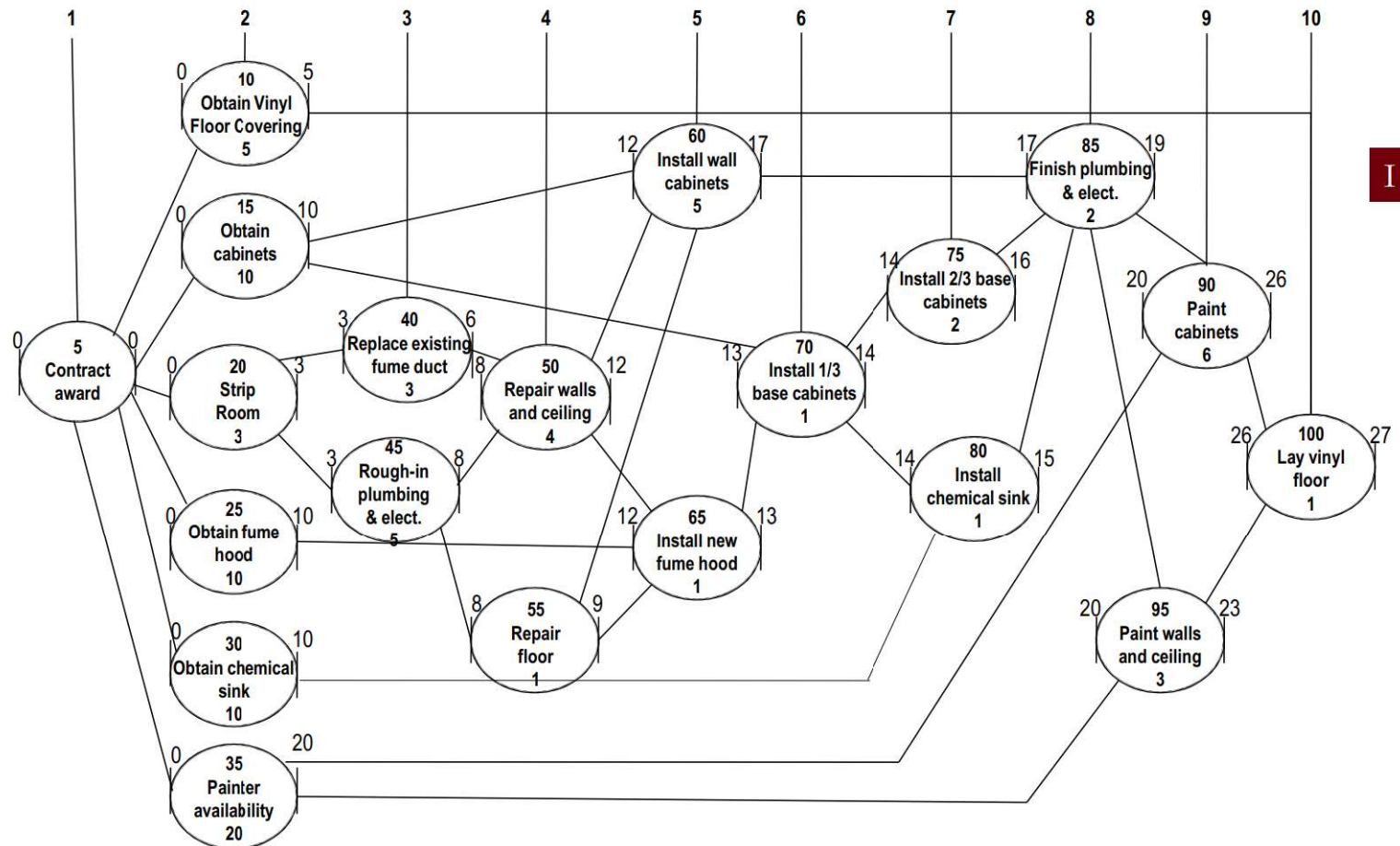
I

1. If more than one activity with no predecessors, create a dummy start activity O. Assign $ES_0 = 0$
2. Set $EF_0 = ES_0 + T_0$
3. Find an activity j , for which the EF_i for all its immediate predecessors are known. Let P be the set of predecessors of j . If no remaining activities then Stop.
4. Set $ES_j = \max_{i \in P} \{EF_i\}$
5. Set $EF_j = ES_j + T_j$
6. Go to 3.

CHEMICAL LAB EXAMPLE – PRECEDENCE DIAGRAM



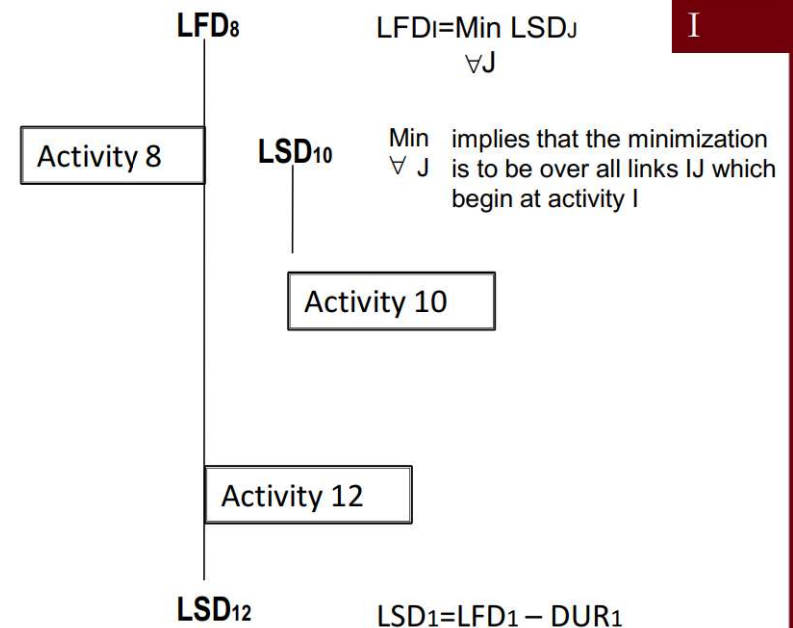
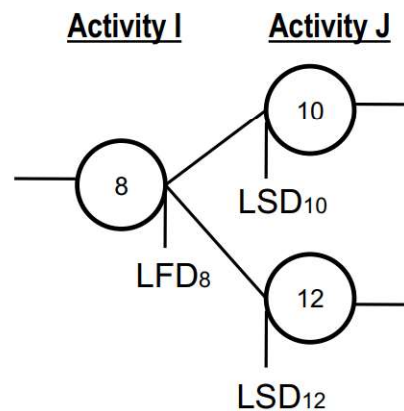
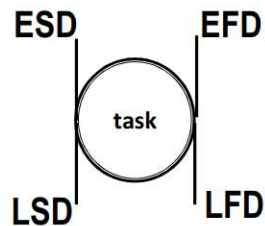
CHEMICAL LAB EXAMPLE – EARLY-START SCHEDULE



NETWORK COMPUTATION

DETERMINATION OF LATE TIMES

- LSD: Late start date
- LFD: Late finish date
- LFD_t : Assigned, usually equal to EFD_t
- $LSD_t = LFD_t - T_t$



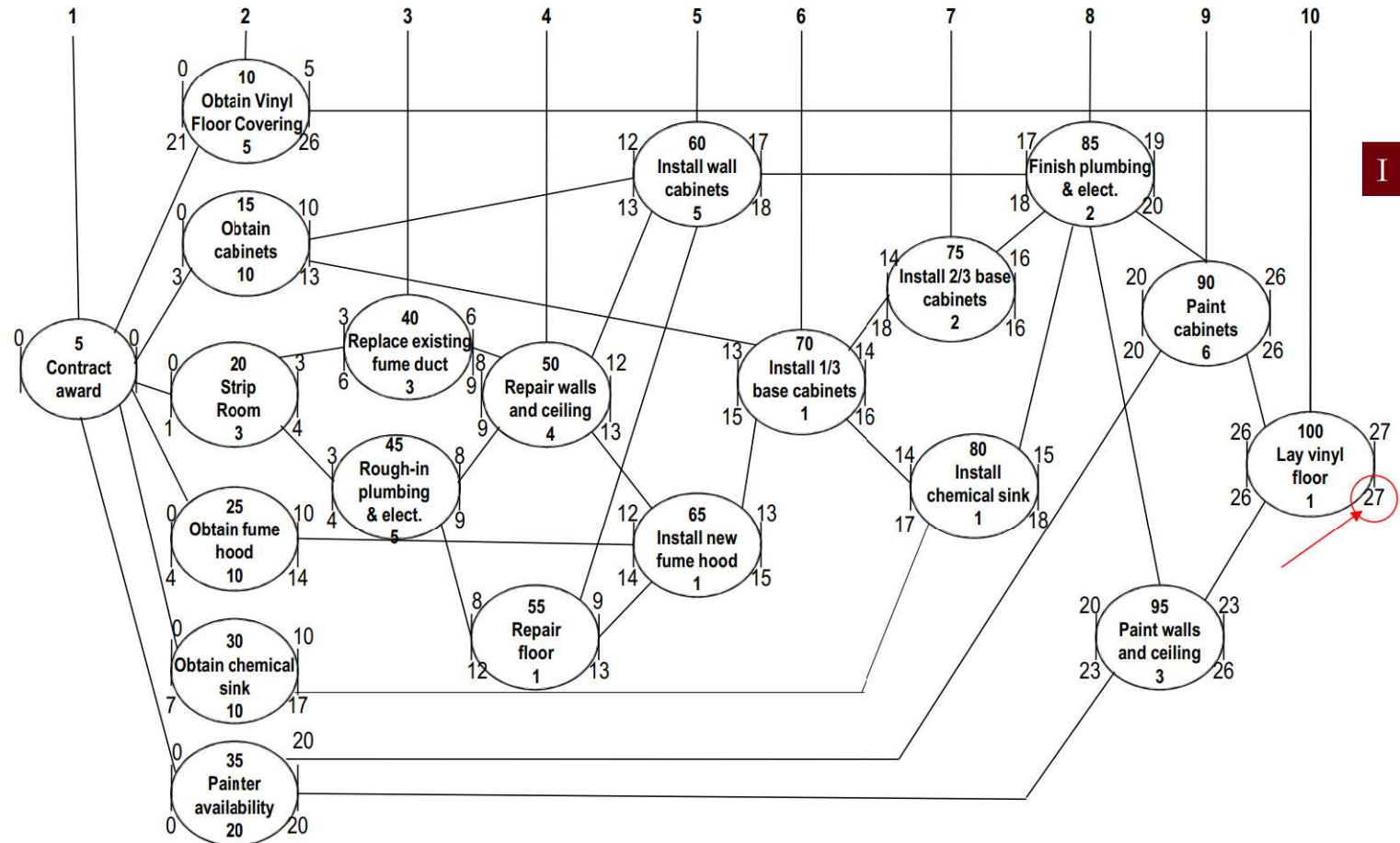
LATE ACTIVITY TIME COMPUTATIONS

► Backward pass computations

I

1. If more than one activity with no successors, create a dummy finish activity *fin*. Assign $LF_{fin} = EF_{fin}$
2. Set $LS_{fin} = LF_{fin} + T_{fin}$
3. Find an activity *i*, for which the LS_j for all its immediate successors are known. Let *S* be the set of successors of *i*. If no remaining activities then Stop.
4. Set $LF_i = \min_{j \in S} \{LS_j\}$
5. Set $LS_i = LF_i + T_i$
6. Go to 3.

CHEMICAL LAB EXAMPLE – LATE-START SCHEDULE

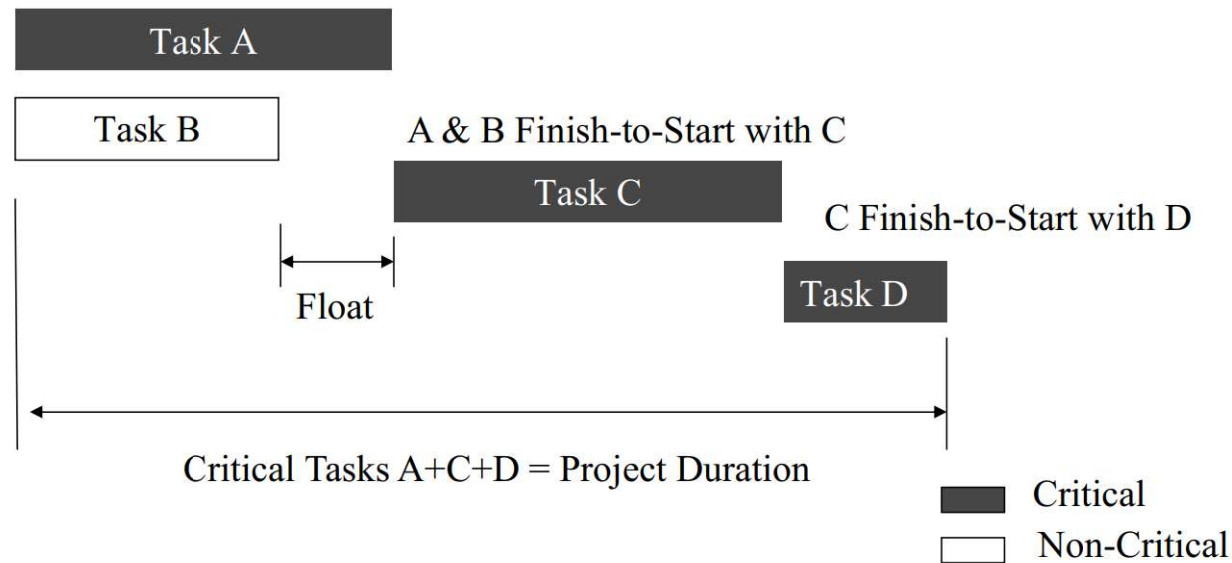


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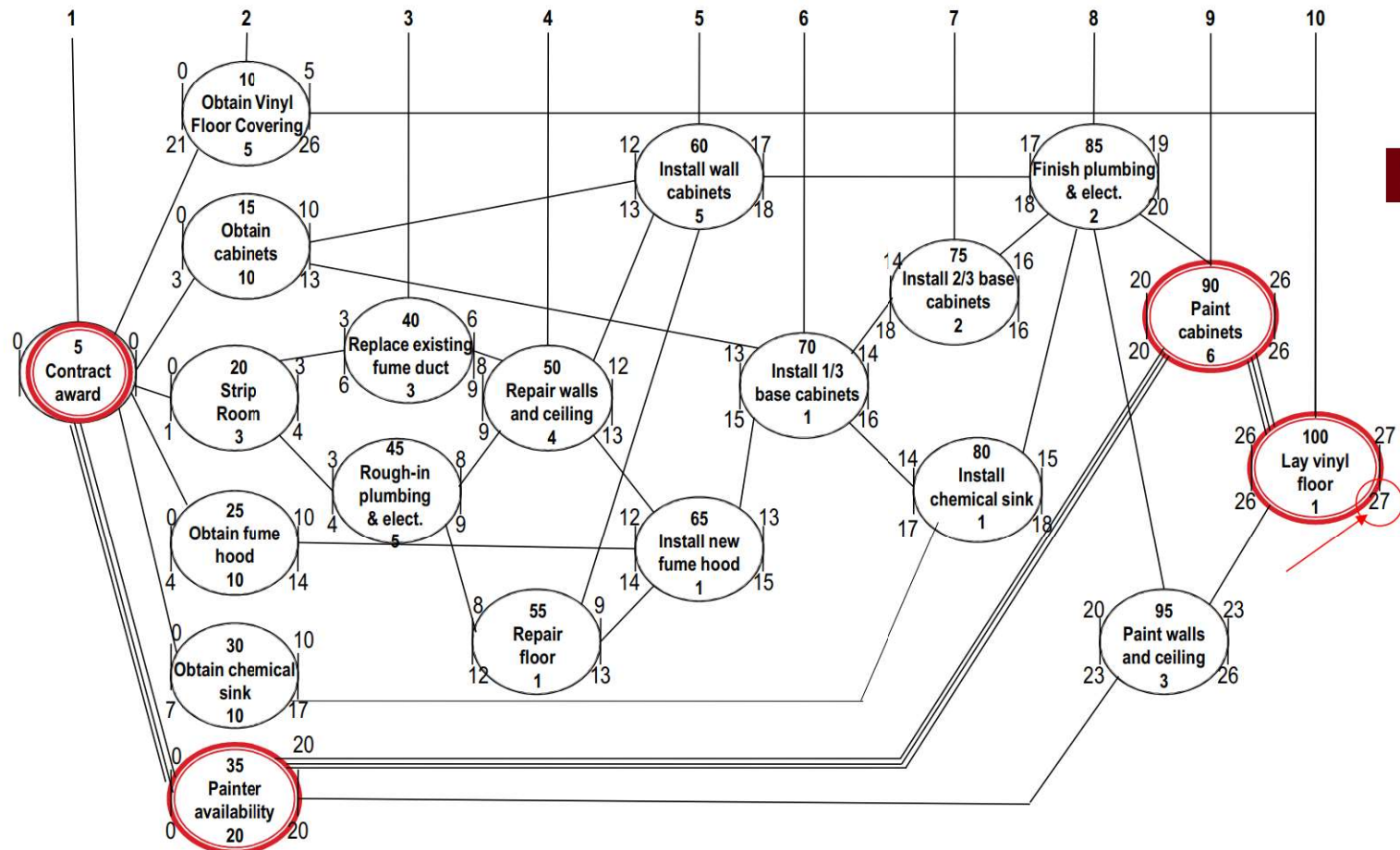
CRITICAL PATH

- A project's critical path is the longest path
- Tasks on the critical path are called critical tasks
- Critical implies the task must occur as scheduled for the project to finish on time
- Non-critical tasks are those tasks not on the critical path

CRITICAL PATH



CHEMICAL LAB EXAMPLE – CRITICAL PATH

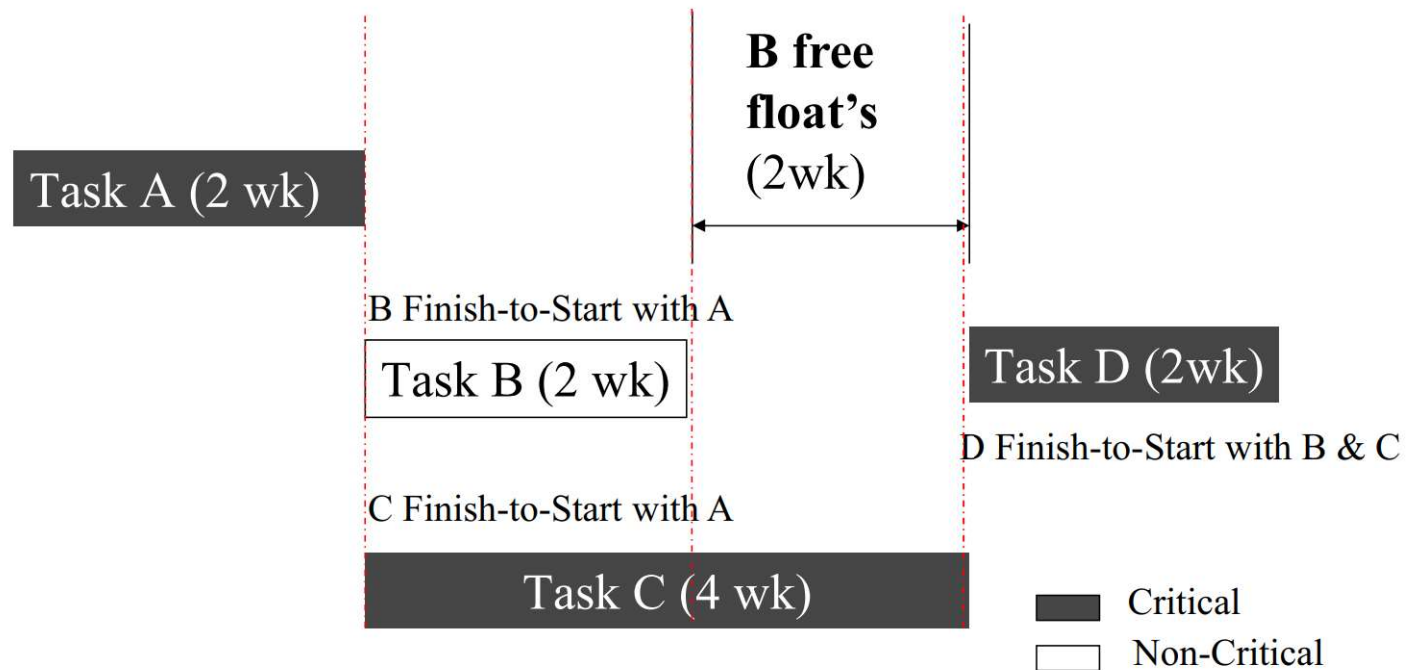


NETWORK COMPUTATION – FLOAT DETERMINATION

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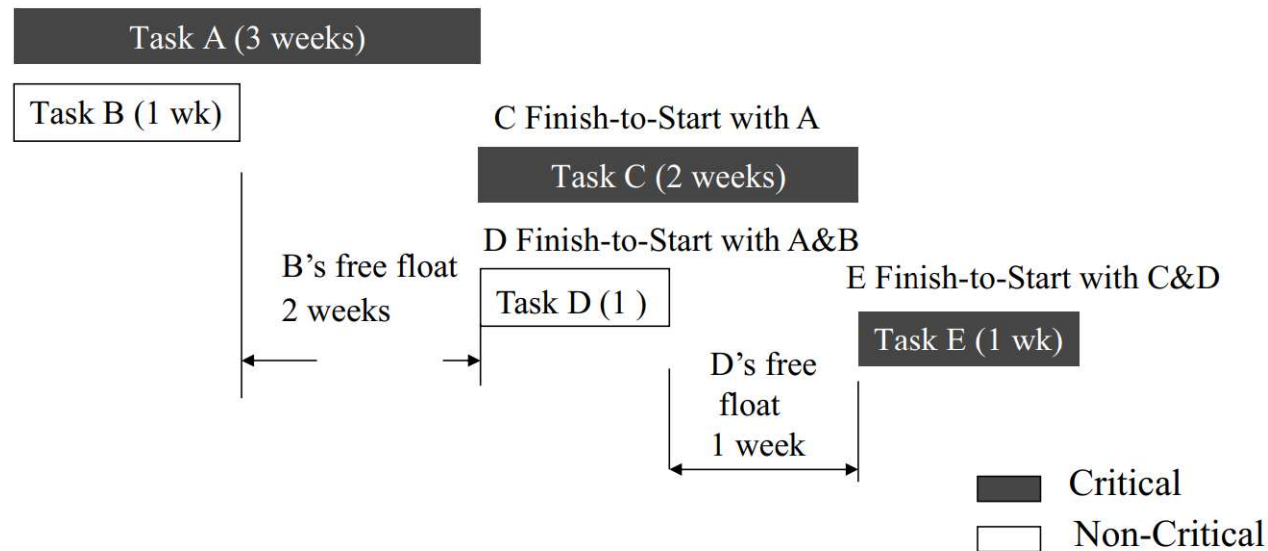
- **Free Float:** Free float represents the amount of time that a schedule activity can be delayed without delaying the early start date of any immediate successor activity within the network path.
- **Total Float:** Total float represents the amount of time an activity can be delayed without delaying the overall project duration and is also called “float” or “slack”.
- **Negative Float:** Negative float, also known as negative slack, is the amount of time beyond a project's scheduled completion that a task within the project requires. Essentially, negative float is the amount of time that must be saved to bring the project to completion on time.

FREE FLOAT



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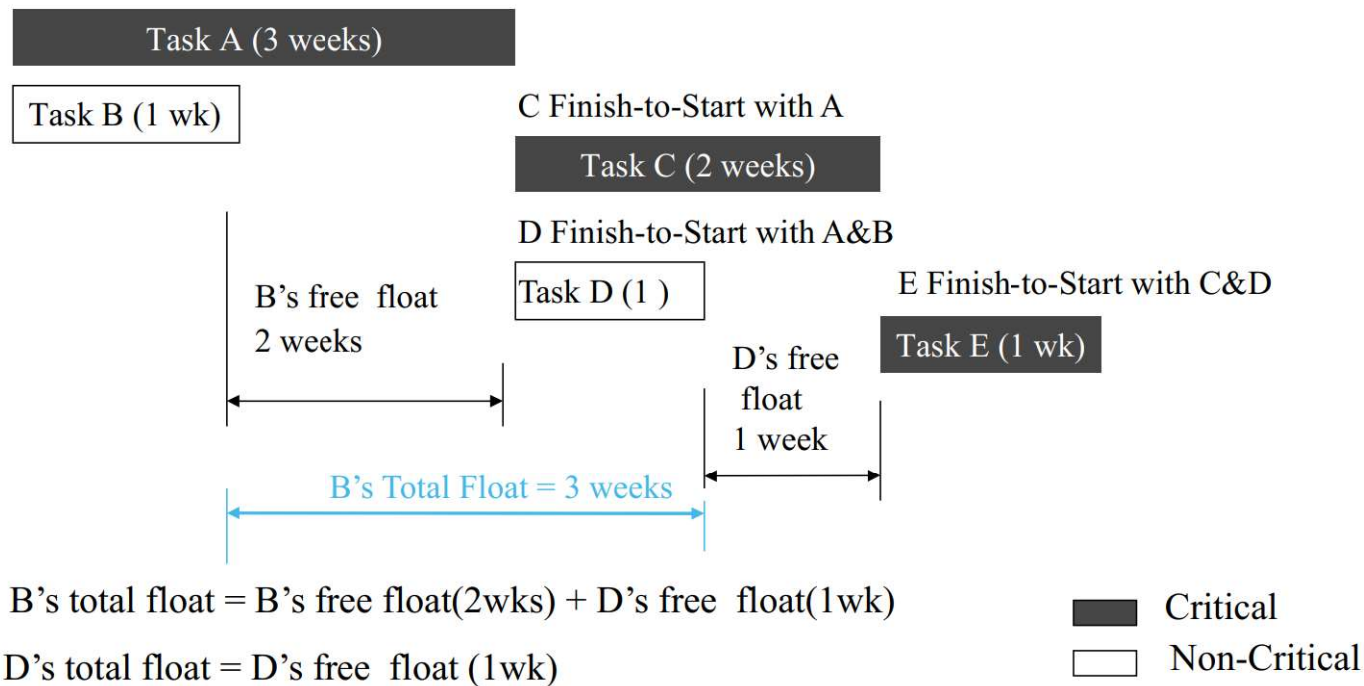
FREE FLOAT



FREE FLOAT

- Non-critical tasks can have Free Float (also called free slack)
- Free Float is the amount of time a task can be delayed without delaying another task early start date
 - ▶ Let $lag_{ij} = ESD_j - EFD_i = ES_j - EF_i$
 - ▶ $FF_i = \min_{j \in S} ESD_j - EFD_i = \min_{j \in S} lag_{ij}$

FREE FLOAT



TOTAL FLOAT

- Non-critical tasks can have **Total Float** (also called total slack time)
- Total Float (total slack) is the amount of time a task can be delayed without delaying the end date of the project

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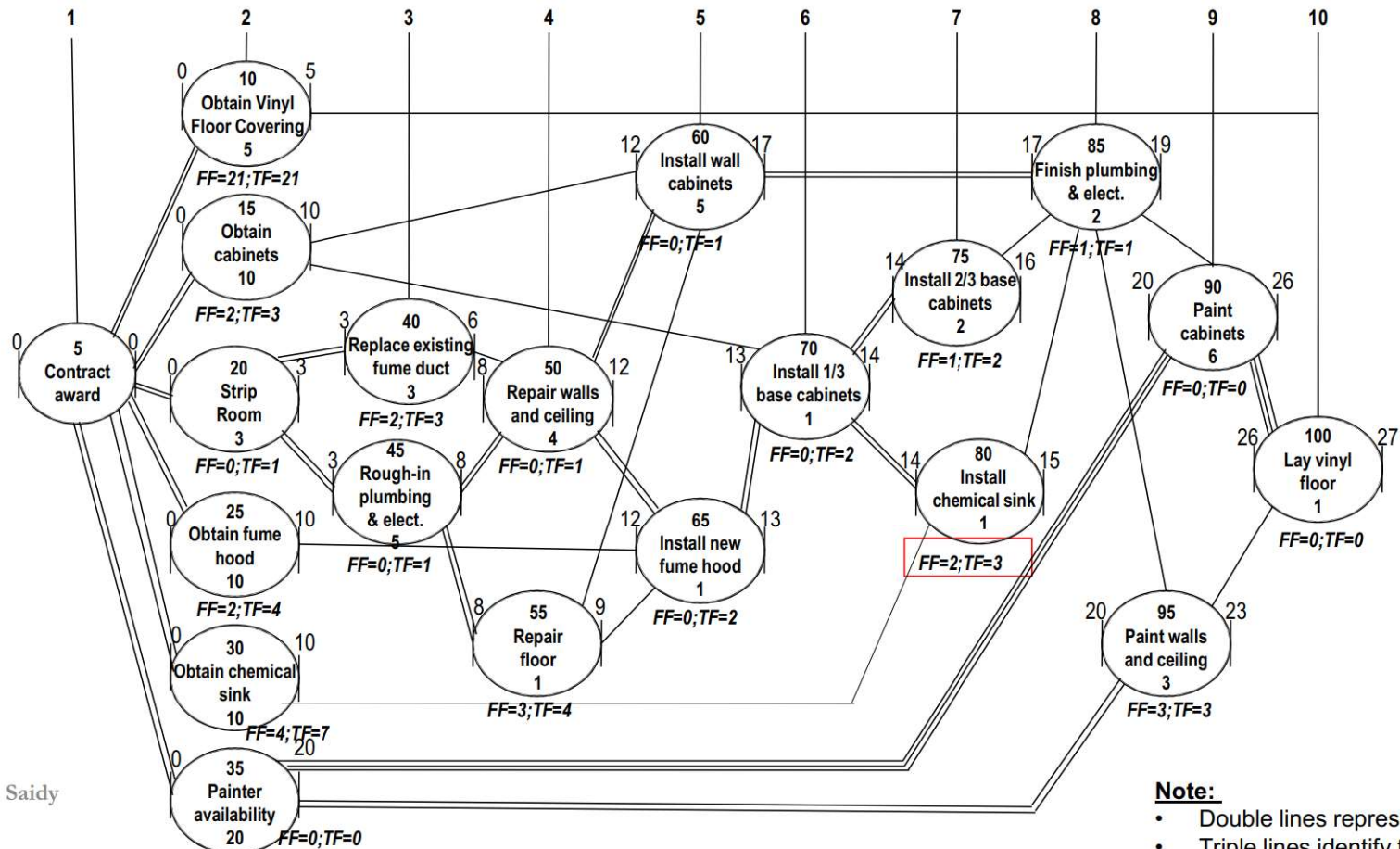
$$\begin{aligned} \blacktriangleright TF_i &= LF_i - EF_i = \min LS_j - EF_i \\ &= \min_{j \in S} (ES_j + TF_j - EF_i) = \min (TF_j + lag_{ij}) \end{aligned}$$

INTF & INDF

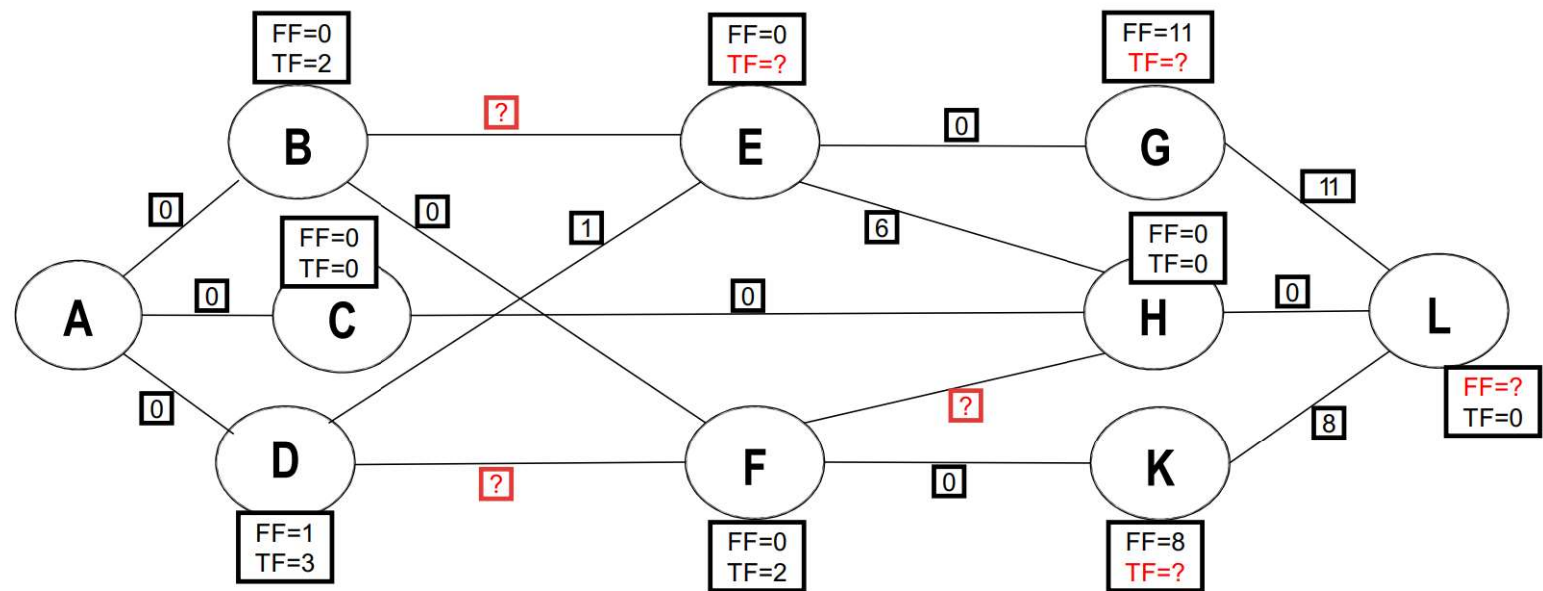
I

- **Interfering Float (INTF):** The amount of time that a schedule activity can be delayed or extended from its early start date without delaying the project finish date, but delaying an activity into interfering float will delay the start of one or more following non-critical activities. If an activity is delayed for the amount of the Free and Interfering Float, its successor activities are critical.
 - Mathematically: $\text{Total Float} - \text{Free Float} = \text{Interfering Float}$
- **Independent Float (INDF):** The maximum amount of time an activity can be delayed without delaying the early start of the succeeding activities and without being affected by the allowable delay of any predecessor activity.
 - Mathematically: $\text{Earliest Successors' Early Start} - \text{Earliest Predecessors' Late Finish} - \text{Activity's duration} = \text{Independent Float}$

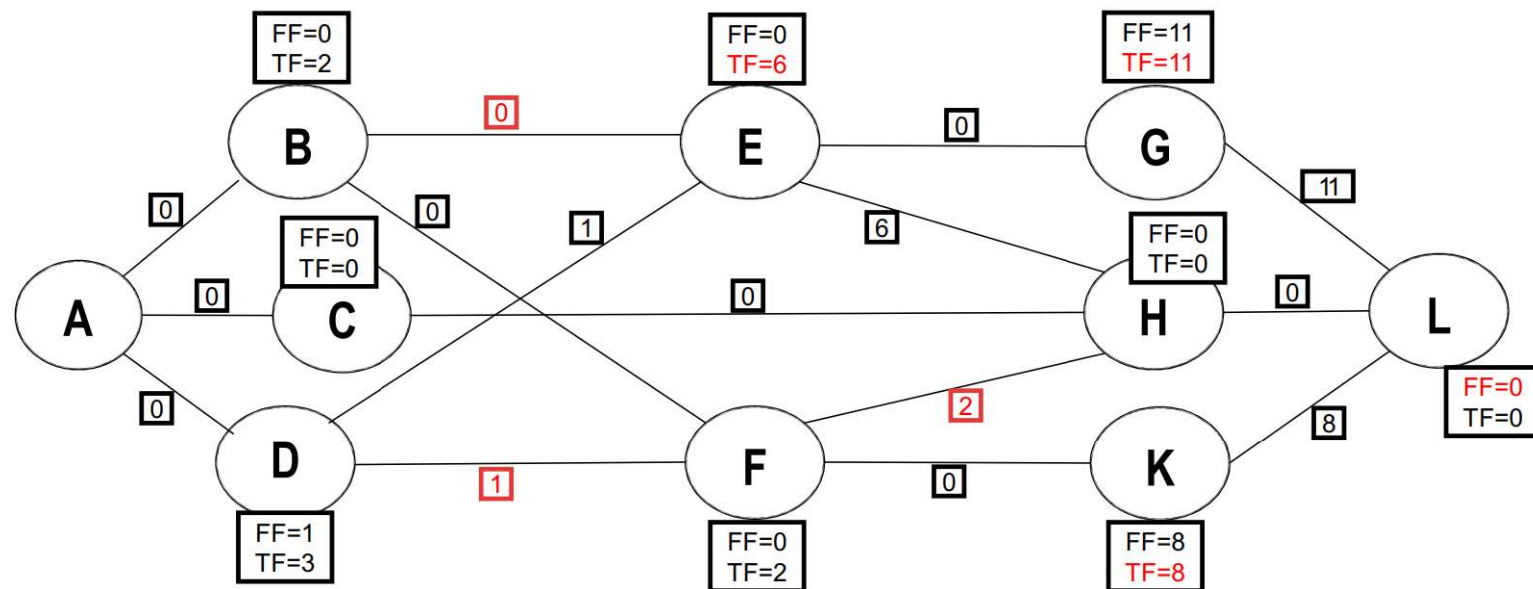
CHEMICAL LAB EXAMPLE – CRITICAL PATH



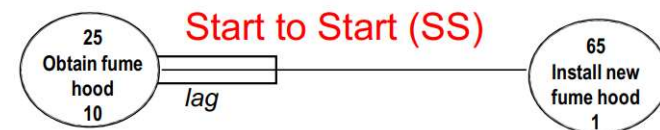
CRITICAL PATH - EXAMPLE



CRITICAL PATH - EXAMPLE



OVERLAPPING NETWORKS – FLOAT COMPUTATIONS



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OVERLAPPING NETWORKS – FLOAT COMPUTATIONS

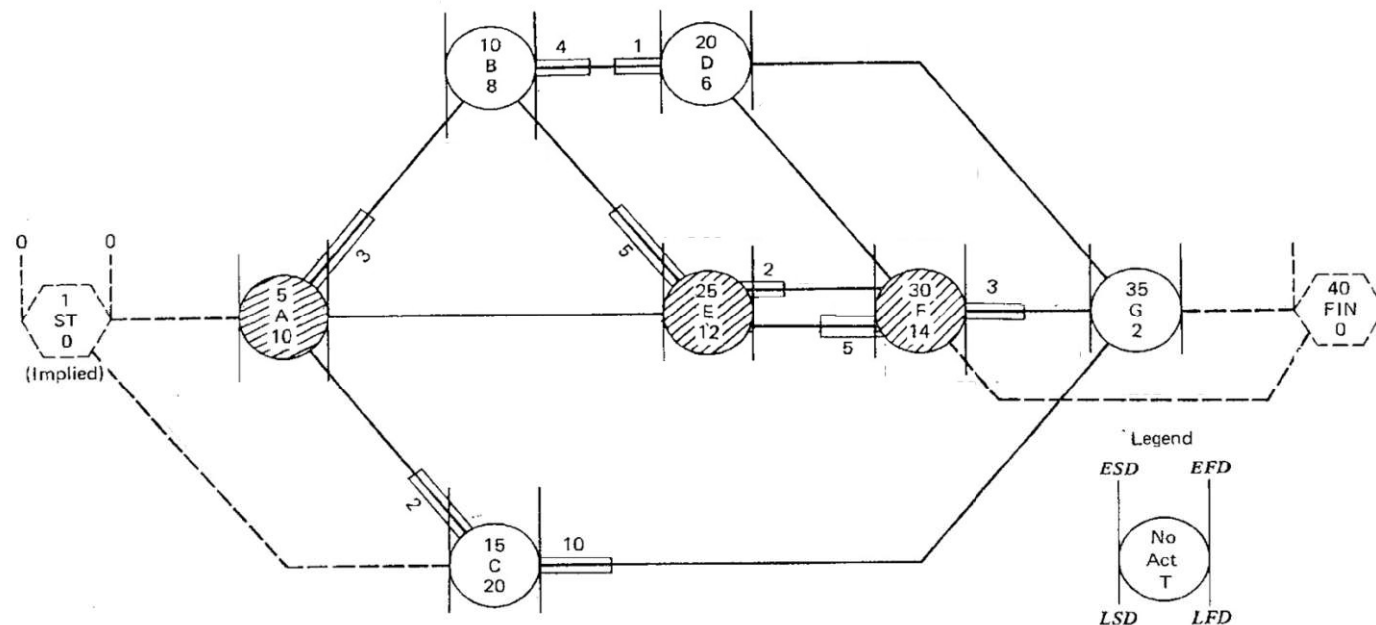
- $TF_i = \min_{j \in S} (TF_j + lag_{ij})$

- $lag_{ij} = \begin{cases} ESD_j - EFD_i - LT & \leftarrow \text{Finish to Start (FS)} \\ ESD_j - ESD_i - LT_i & \leftarrow \text{Start to Start (SS)} \\ EFD_j - EFD_i - LT_j & \leftarrow \text{Finish to Finish (FF)} \\ EFD_j - ESD_i - LT_i - LT_j & \leftarrow \text{Start to Finish (SF)} \end{cases}$

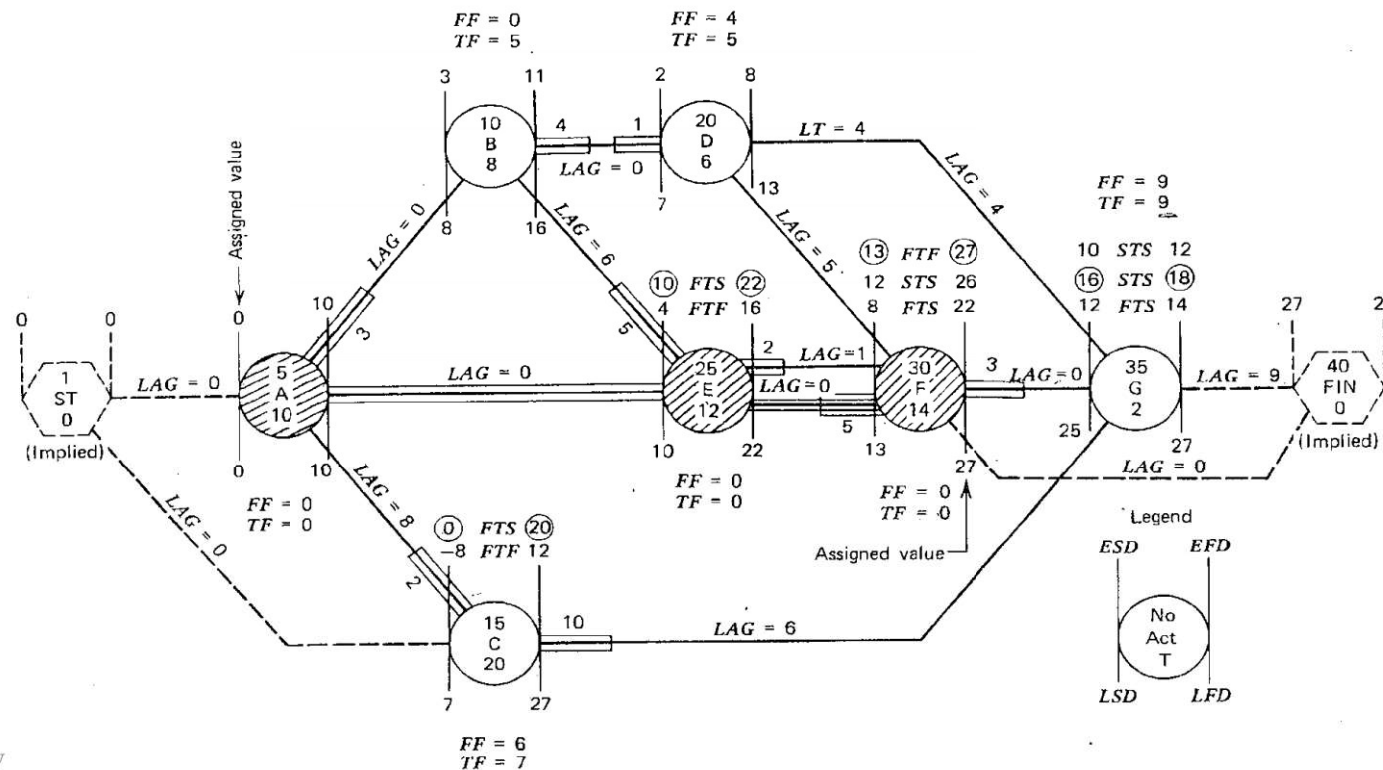
- $FF_i = \min_{j \in S} (lag_{ij})$

SCHEDULE COMPUTATIONS – OVERLAPPING NETWORKS

I



SCHEDULE COMPUTATIONS – OVERLAPPING NETWORKS



Program Evaluation Review Technique (PERT)

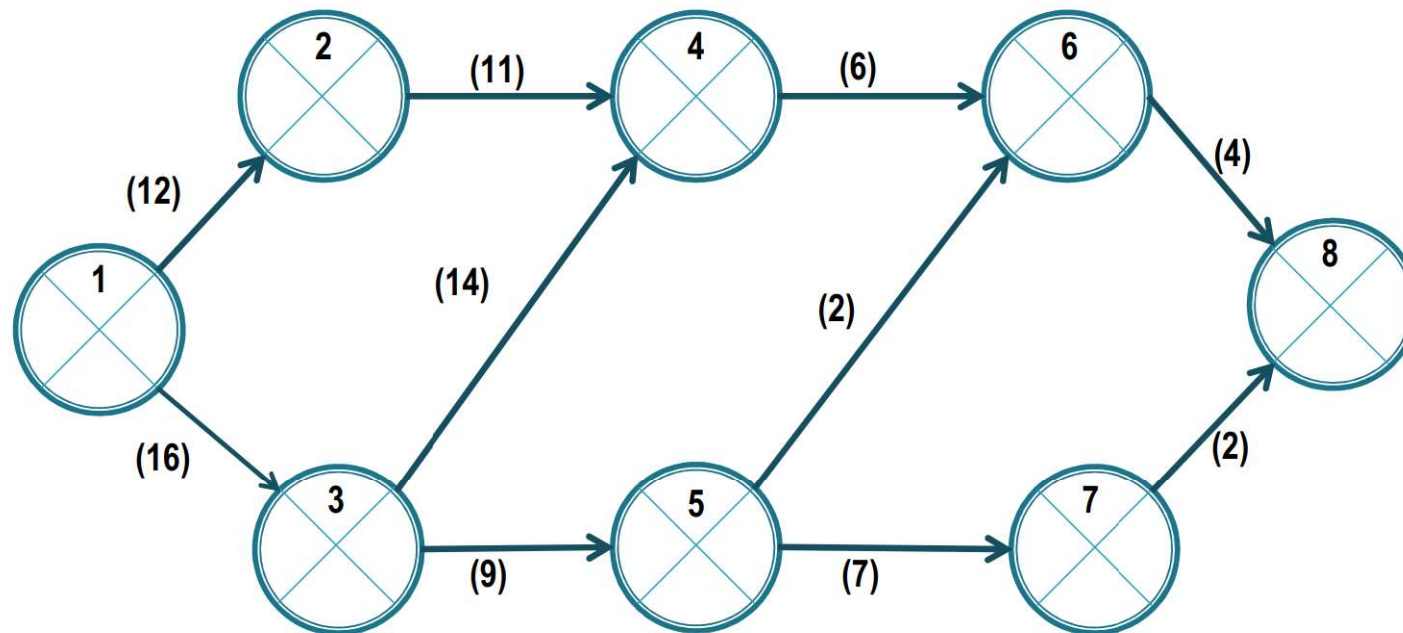
Project Management – Project Scheduling

PERT – PROGRAM EVALUATION REVIEW TECHNIQUE

II

- Developed for the Navy by Booz Allen and Hamilton
- Mainly used for R&D Projects to control time aspect of projects
- Uses Probabilistic Time Estimates to aid in determining the probability that project could be completed by some given date.

AOA NETWORK OF ACTIVITIES



- *TE* = earliest event occurrence time
- *TL* = latest allowable event occurrence time
- *ES* = Event Slack



EVENT TIME COMPUTATION

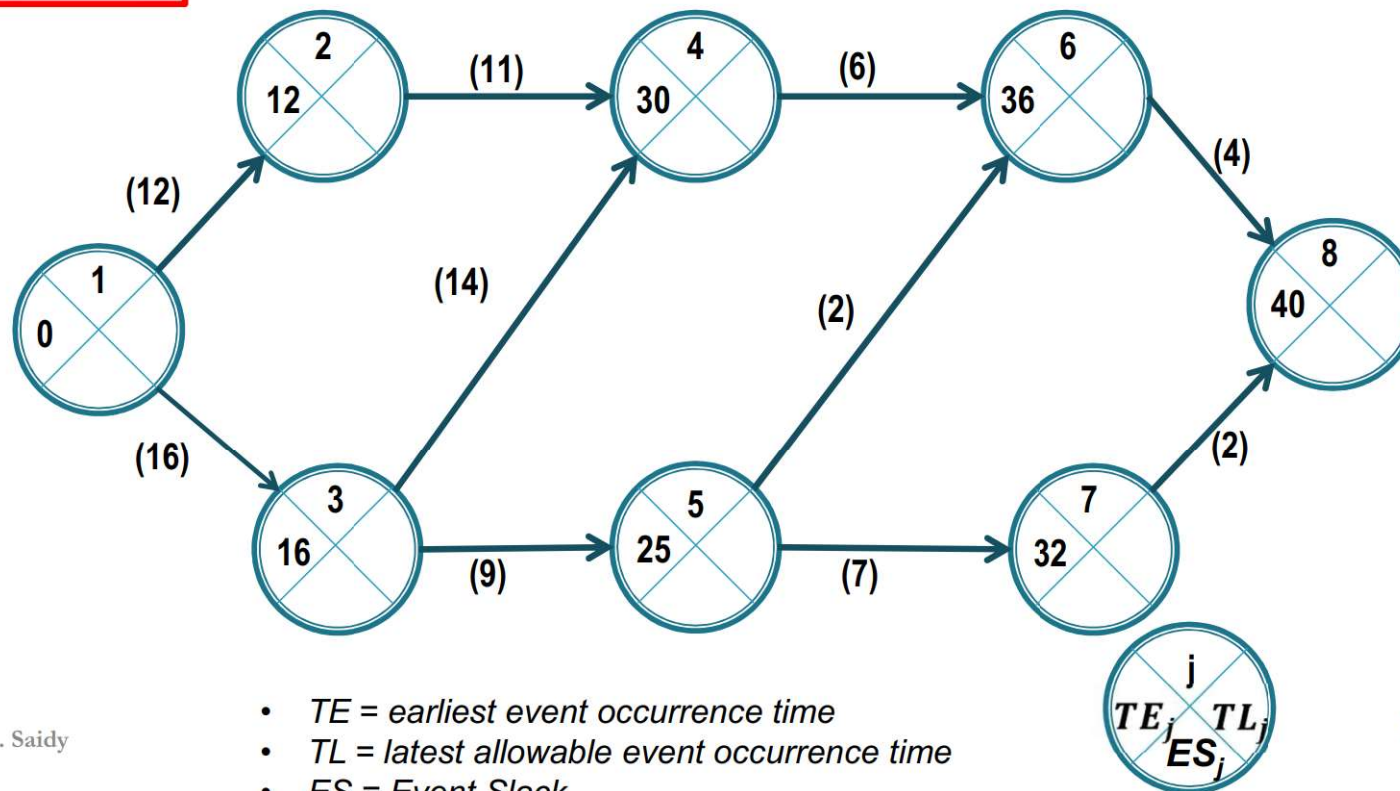
► Forward pass computations

Number nodes in such a way so that $i < j \forall \text{ act-}ij$

1. Starting at node 0, Assign $TE_0 = 0$
2. Find a node $j, j > i$ and for which all immediately preceding nodes TE_i are known. If no node is found then Stop. Let P denote the set of arcs ending at node j .
3. Set $TE_j = \max_{i \in P} \{TE_i + t_{ij}\}$
4. Go to 2.

AOA NETWORK OF ACTIVITIES

$$TE_j = \max_{i \in P} \{TE_i + t_{ij}\}$$



II

- *TE* = earliest event occurrence time
- *TL* = latest allowable event occurrence time
- *ES* = Event Slack

EVENT TIME COMPUTATION

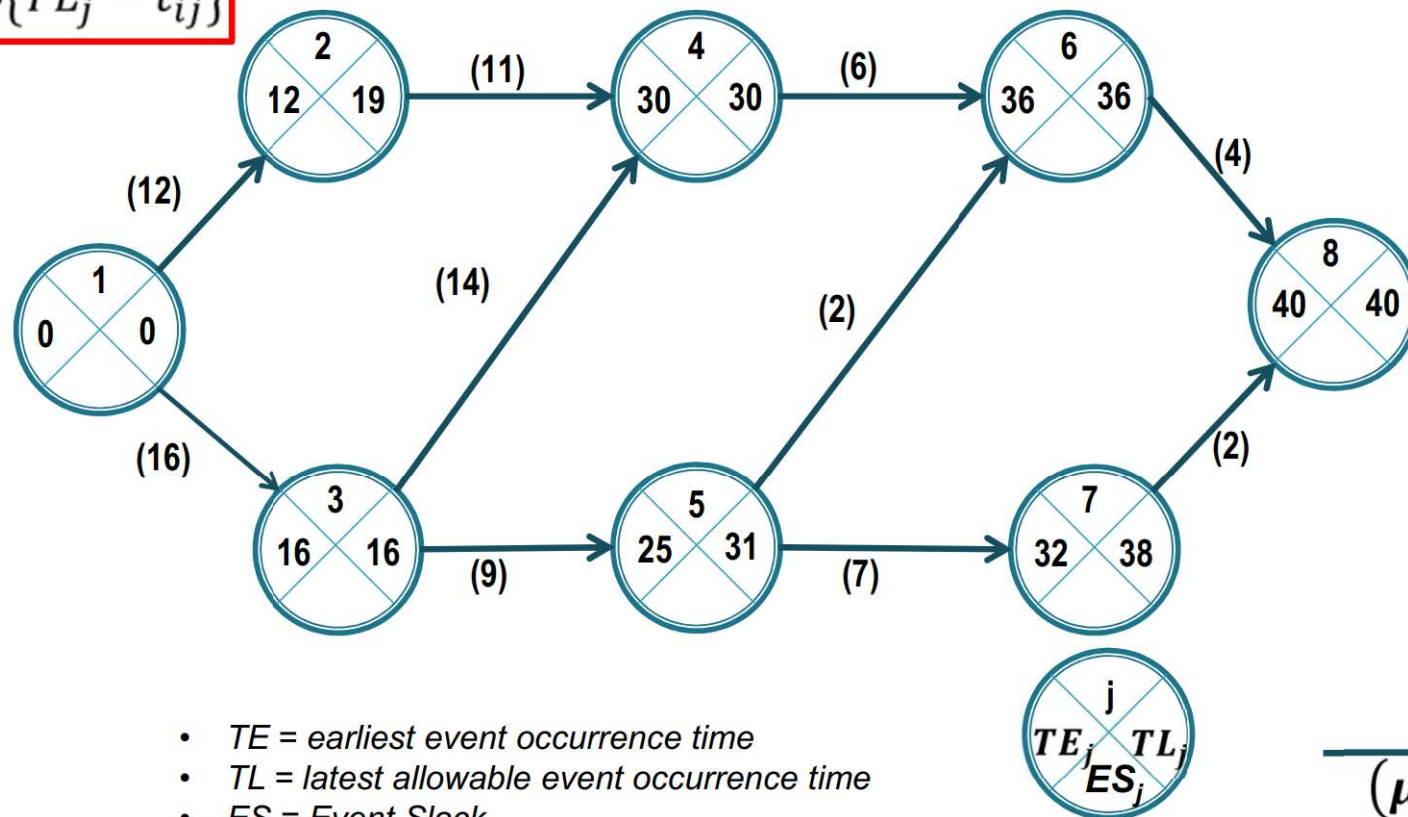
► Backward pass computations

Let the highest numbered node be labeled *fin*

1. Assign $TL_{fin} = TE_{fin}$
2. Find a node i , $i < j$ and for which all immediately succeeding nodes TL_j are known. Let S be the set of arcs starting at node i . Set $TL_i = \min_{j \in S} \{TL_j - t_{ij}\}$
3. Go to 2.

AOA NETWORK OF ACTIVITIES

$$TL_i = \min_{j \in S} \{TL_j - t_{ij}\}$$



PERT COMPUTATIONS USING MEAN VALUES

II

▶ Computing Activity Early Start and Finish Dates

- $ESD_{ij} = TE_i$
- $EFD_{ij} = TE_i + t_{ij}$

▶ Computing Activity Late Start and Finish Dates

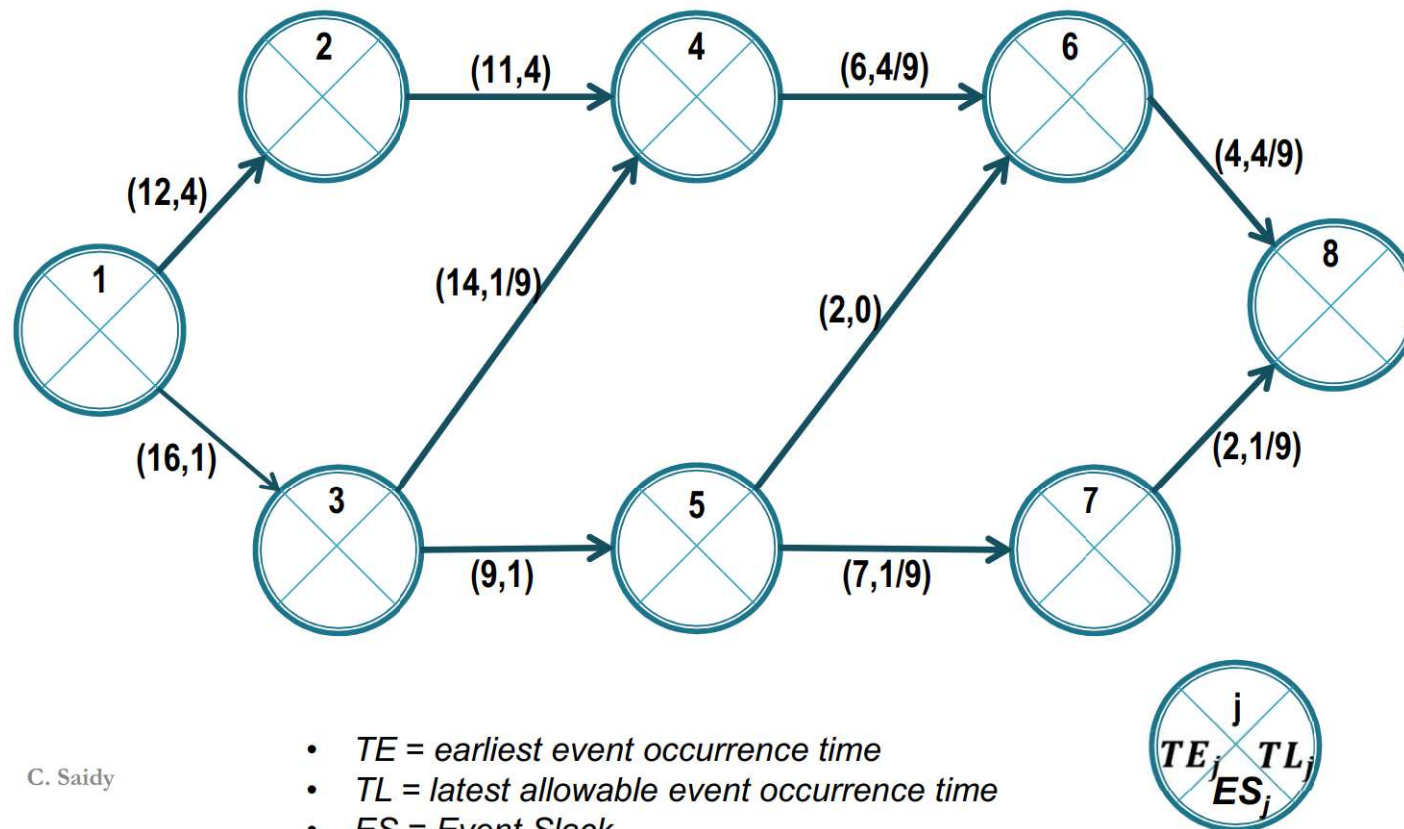
- $LFD_{ij} = TL_j$
- $LSD_{ij} = TL_j - t_{ij}$

▶ Computing Event Slack

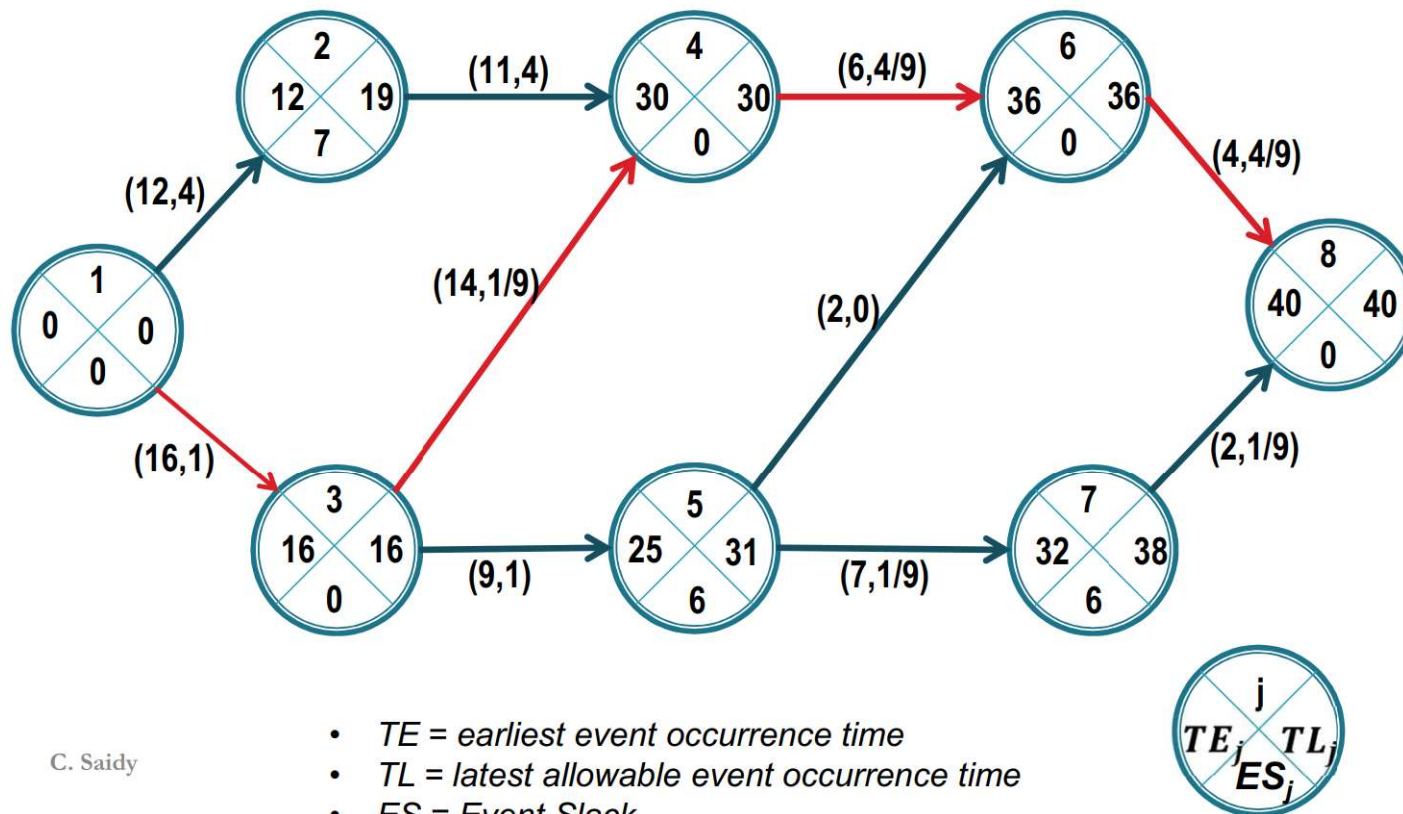
- $ES_i = TL_i - TE_i$

$$\text{Activity Total Slack(float)} = ATSi_j = \begin{cases} TL_j - TE_i - t_{ij} \\ LS_{ij} - ES_{ij} \\ LF_{ij} - EF_{ij} \end{cases}$$

PERT EXAMPLE



PERT EXAMPLE



- TE = earliest event occurrence time
- TL = latest allowable event occurrence time
- ES = Event Slack

ACTIVITY DURATIONS IN PERT NETWORK

II

- Beta Distribution is used to model activity durations
- Mean and variance of the beta distribution are approximated by the following equations

$$\mu = \frac{a + 4m + b}{6}$$

$$\sigma = \frac{b - a}{6}$$

NETWORK COMPUTATION

II

- The length of any path in a PERT network is the sum of the durations of activities along that path
- It is assumed that the number of activities along a path is large enough to make the Central Limit Theorem applicable
- The CLT states that the sum of n independently distributed random variables is a random variable that is normally distributed with mean equal to the sum of the mean durations of activities along that path and with variance equal to the sum of the variances.

Central Limit Theorem (for large “n”)

- If $X \sim N(\mu_i, \sigma_i^2)$

$$\text{Let } Y = X_1 + X_2 + \dots + X_n$$

$$\rightarrow Y \sim N(\mu_y, \sigma_y^2)$$

Where:

$$\mu_y = \sum_{i=1}^n \mu_i$$

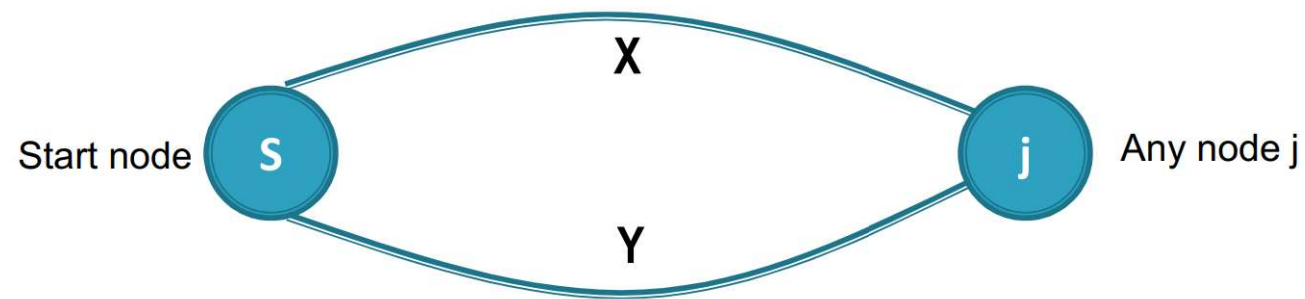
$$\sigma_y^2 = \sum_{i=1}^n \sigma_i^2$$

For that reason:

$$TE_i \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$$

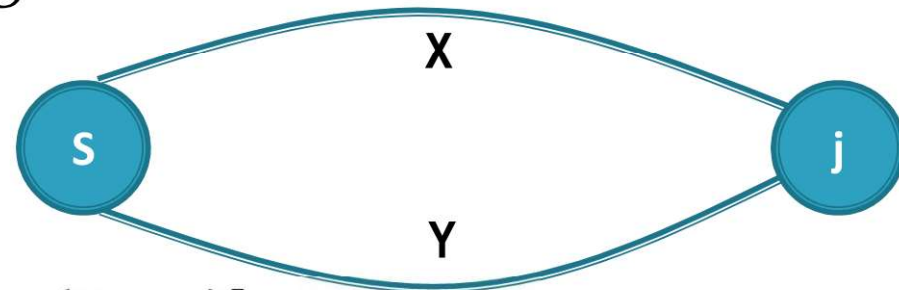
Belonging to the longest path

PERT MERGE BIAS



- ▶ Consider the case where 2 paths exists from the start node to some node j in the network
- ▶ X and Y are random variables representing the durations of theses paths
- ▶ Let Z represent TE_j , early event time of event j

PERT MERGE BIAS

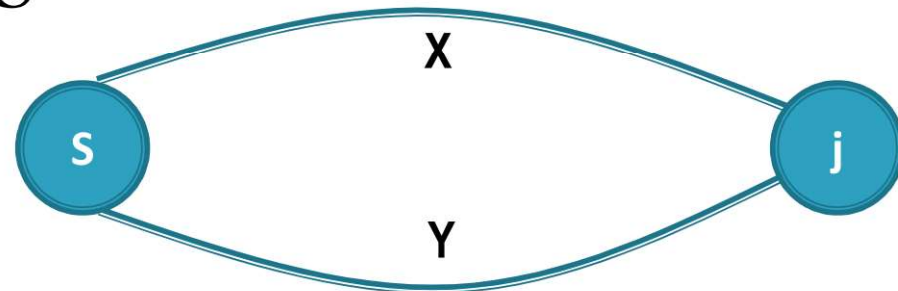


- $$F_Z(a) = P(Z \leq a) = P[(X \leq a) \cap (Y \leq a)]$$

$$= \underline{P(Y \leq a | X \leq a) P(X \leq a)} \rightarrow \text{also} = P(X \leq b | Y \leq a) * P(Y \leq a)$$
- If X and Y are independent then $F_Z(a) = F_Y(a) F_X(a)$
- In general X & Y will share common resources which will make them positively correlated, i.e.,

$$P(Y \leq a | X \leq a) \geq P(Y \leq a)$$

PERT MERGE BIAS



II

- ▶ $F_Z(a) = P(Z \leq a) \geq P(Y \leq a) P(X \leq a)$
- ▶ If $\rho_{xy} \rightarrow 1$ then $F_Z(a) \rightarrow \text{Min}\{F_Y(a), F_X(a)\}$
- ▶ Hence, $\underline{F_Y(a)F_X(a)} \leq F_Z(a) \leq \underline{\text{Min}\{F_Y(a), F_X(a)\}}$

Lower bound LB
(when X & Y are independent)

Upper bound UB
(when X & Y are dependent)

PERT MERGE BIAS

II

- Bias because of merging paths could be ignored in practice if longest path is 3σ greater than any other path in the network
- This bias increase as the standard deviation of the merging paths increases

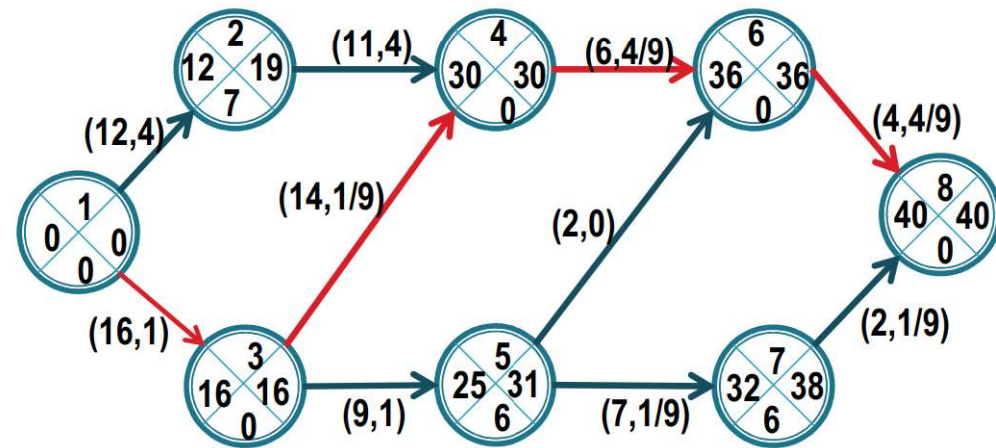
PERT EXAMPLE

- Consider Event #4:
→ What is the probability of $TE_4 \leq 30$?

Reminder:

$$P(X \leq a) = P\left(z \leq \frac{a - \mu}{\sigma_X}\right)$$

Then look the probability in the z-table



Path (t)	Activities	Mean	Variance	P($TE_4 \leq 30$)

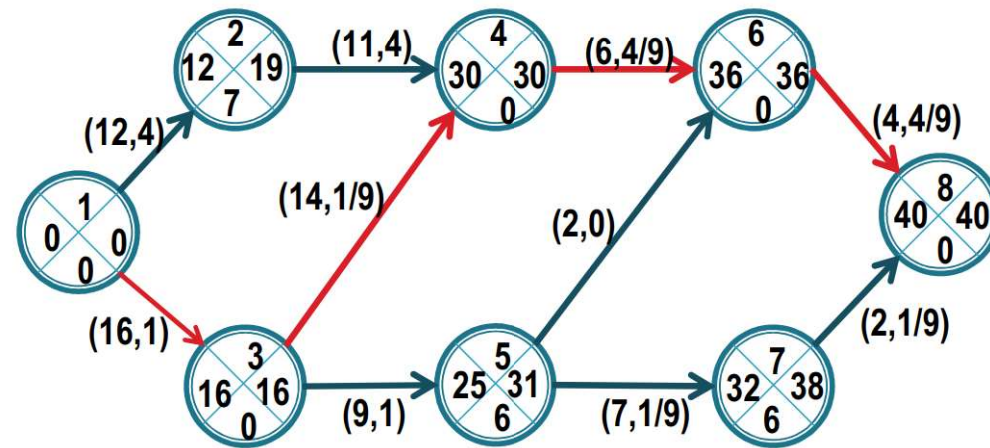
PERT EXAMPLE

- Consider Event #4:
→ What is the probability of $TE_4 \leq 30$?

Reminder:

$$P(X \leq a) = P\left(z \leq \frac{a - \mu}{\sigma_X}\right)$$

Then look the probability in the z-table



Path (t)	Activities	Mean	Variance	$P(TE_4 \leq 30)$
X	1-2-4	23	8	$P\left(z \leq \frac{30-23}{\sqrt{8}}\right) = 0.9934$
Y	1-3-4	30	10/9	$P\left(z \leq \frac{30-30}{\sqrt{10/9}}\right) = 0.5$

$$F_Y(a)F_X(a) \leq F_Z(a) \leq \min\{F_Y(a), F_X(a)\}$$

$$\rightarrow 0.9934 * 0.5 \leq P(TE_4 \leq 30) \leq \min(0.5, 0.9934)$$

$$\rightarrow 0.4967 \leq P(TE_4 \leq 30) \leq 0.5$$

PERT EXAMPLE

$$P\left(z \leq \frac{30-2}{\sqrt{8}}\right) = P(z \leq 2.48) = 0.9934$$

$$P\left(z \leq \frac{30-30}{\sqrt{10/9}}\right) = P(z \leq 0) = 0.5$$

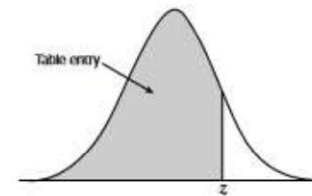


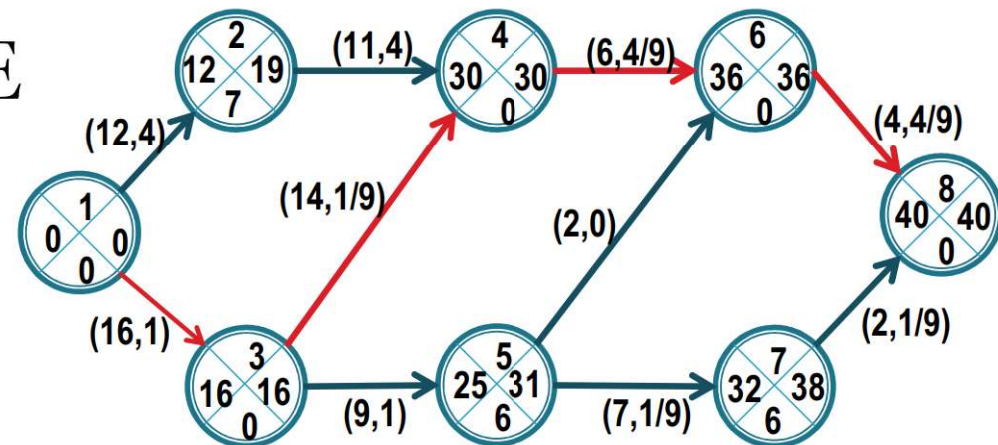
Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

II

PERT EXAMPLE

What is the probability of the Project ending before 40 days?

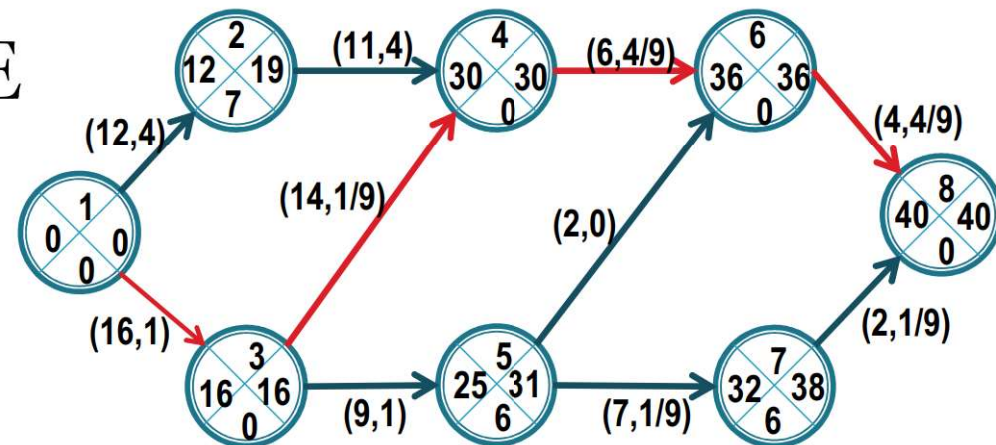


II

Path (t)	Activities	Mean	Variance	$P(t \leq 40)$

PERT EXAMPLE

What is the probability of the Project ending before 40 days?

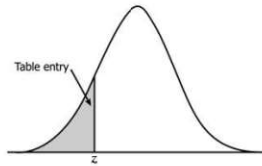


Path (t)	Activities	Mean	Variance	$P(t \leq 40)$
X	1-3-4-6-8	40	2	$P\left(z \leq \frac{40 - 40}{\sqrt{2}}\right) = 0.5$
Y	1-2-4-6-8	33	$88/9$	$P\left(z \leq \frac{40 - 33}{\sqrt{88/9}}\right) = 0.9906$
Z	1-3-5-6-8	31	$24/9$	Events Z and W can be neglected since: $31 + 3 \cdot \text{Sqrt}(24/9) = 35.7 < 40$ $34 + 3 \cdot \text{Sqrt}(22/9) = 38.47 < 40$
W	1-3-5-7-8	34	$22/9$	

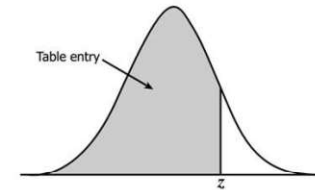
$$F_Y(a)F_X(a) \leq F_Z(a) \leq \text{Min}\{F_Y(a), F_X(a)\}$$

$$\rightarrow 0.9906 * 0.5 \leq P(t \leq 40) \leq \text{Min}(0.5, 0.9906)$$

$$\rightarrow 0.4953 \leq P(t \leq 40) \leq 0.5$$



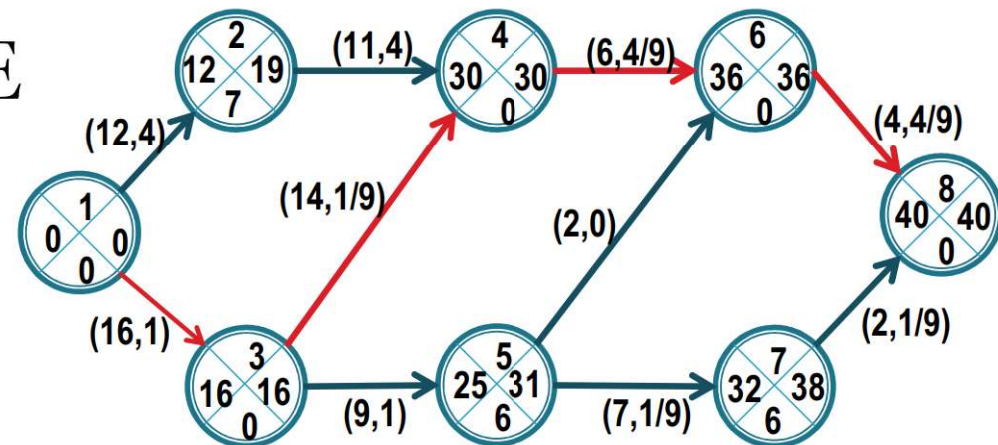
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0022	.0021	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

PERT EXAMPLE

What is the probability of the Project ending before 38 days?

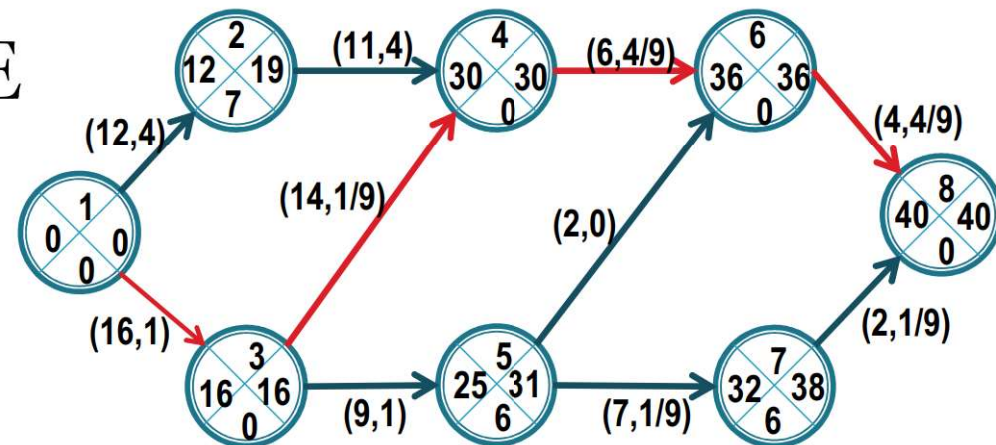


II

Path (t)	Activities	Mean	Variance	$P(t \leq 38)$
X	1-3-4-6-8	40	2	
Y	1-2-4-6-8	33	$88/9$	
Z	1-3-5-6-8	31	$24/9$	
W	1-3-5-7-8	34	$22/9$	

PERT EXAMPLE

What is the probability of the Project ending before 38 days?

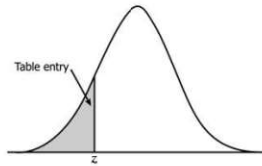


Path (t)	Activities	Mean	Variance	$P(t \leq 38)$
X	1-3-4-6-8	40	2	$P\left(z \leq \frac{38 - 40}{\sqrt{2}}\right) = 0.0793$
Y	1-2-4-6-8	33	$88/9$	$P\left(z \leq \frac{38 - 33}{\sqrt{88/9}}\right) = 0.9535$
Z	1-3-5-6-8	31	$24/9$	Event Z can be neglected since: $31 + 3 \cdot \sqrt{24/9} = 35.7 < 38$
W	1-3-5-7-8	34	$22/9$	$P\left(z \leq \frac{38 - 34}{\sqrt{22/9}}\right) = 0.9963$

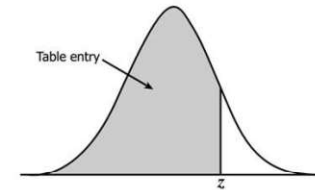
$$F_Y(a)F_X(a) \leq F_Z(a) \leq \min\{F_Y(a), F_X(a)\}$$

$$\rightarrow 0.079 * 0.953 * 0.996 \leq P(t \leq 38) \leq \min(0.079, 0.953, 0.996)$$

$$\rightarrow 0.0753 \leq P(t \leq 38) \leq 0.0793$$

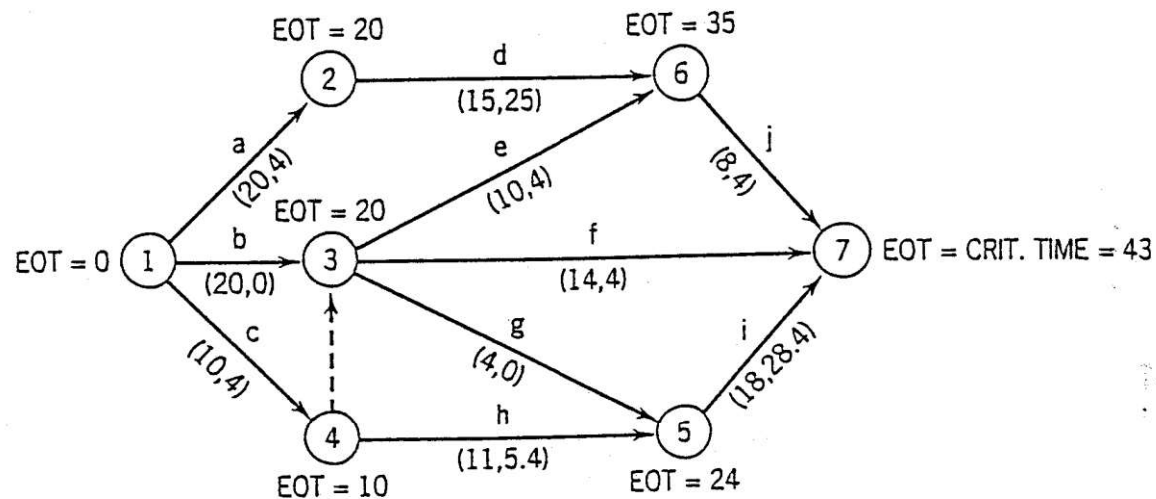


z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0022	.0021	.0021	.0020	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

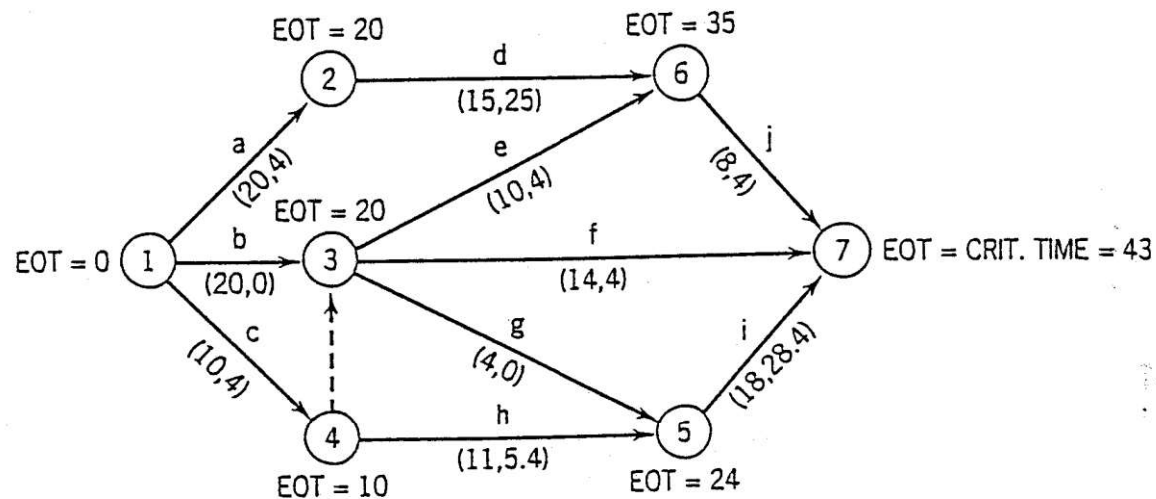
PERT EXAMPLE



- EOT: Earliest Occurrence Time = **TE**
- LOT: Latest Occurrence Time = **TL**
- LST: Latest Starting Time = **LSD**
- EST: Earliest Starting Time = **ESD**

Event	LOT	EOT	Slack
1			
2			
3			
4			
5			
6			
7			
Activity	LST	EST	Slack
a			
b			
c			
d			
e			
f			
g			
h			
i			
j			

PERT EXAMPLE



- EOT: Earliest Occurrence Time = **TE**
- LOT: Latest Occurrence Time = **TL**
- LST: Latest Starting Time = **LSD**
- EST: Earliest Starting Time = **ESD**

Event	LOT	EOT	Slack
1	0	0	0
2	20	20	0
3	21	20	1
4	14	10	4
5	25	24	1
6	35	35	0
7	43	43	0
Activity	LST	EST	Slack
a	0	0	0
b	1	0	1
c	4	0	5
d	20	20	0
e	25	20	5
f	29	20	9
g	21	20	1
h	14	10	4
i	25	24	1
j	35	35	0

DIFFERENCE BETWEEN PERT AND CPM

- **The Critical Path Method (CPM)** is best used for projects that involve activities whose time estimate can be predicted accurately (uncertainties about duration estimates are limited). Example of such activities would include, activities for which the project team has experience or projects for which industry standards around time estimates exist.
- **The PERT Method** is best used for projects that have inherently high degree of uncertainties. Research based projects and application (software) development projects fall in this category .

BASIS FOR COMPARISON	PERT	CPM
Meaning	PERT is a project management technique, used to manage uncertain activities of a project.	CPM is a statistical technique of project management that manages well defined activities of a project.
What is it?	A technique of planning and control of time.	A method to control cost and time.
Orientation	Event-oriented	Activity-oriented
Evolution	Evolved as Research & Development project	Evolved as Construction project
Model	*Probabilistic Model	*Deterministic Model
Focuses on	Time	Time-cost trade-off
Estimates	Three-time estimates	One-time estimate
Appropriate for	High precision time estimate	Reasonable time estimate
Management of Nature of jobs	Unpredictable Activities Non-repetitive nature	Predictable activities Repetitive nature
Critical and Non-critical activities	No differentiation	Differentiated
Suitable for	Research and Development Project	Non-research projects like civil construction, ship building etc.
Crashing concept	Not Applicable	Applicable

SOURCE: Difference Between PERT and CPM: <https://keydifferences.com/difference-between-pert-and-cpm.html>

Schedule Analysis

Project Management – Project Scheduling

SCHEDULE ANALYSIS

- **Time Constrained** – Meet a specified end date
- **Cost Constrained** – Meet a specified Budget
- **Resource Constrained** – Finish without exceeding some specific level of resource usage
- **Resource Leveling** – Even out demand for resources

STRATEGIES FOR MEETING A SPECIFIED END DATE

III

- Parallel processing/ Altering Precedence Relationships
 - Start painting the walls and ceiling before the cabinets are in place and before the plumbers and electricians have completed their work.
 - Painters will have one day to touch up around fixtures and cabinets after their installation.
- New technology/ Expert Resources/ Different method
 - Use pre-painted cabinets. This will reduce duration of paint activity.

STRATEGIES FOR MEETING A SPECIFIED END DATE (CONT'D)

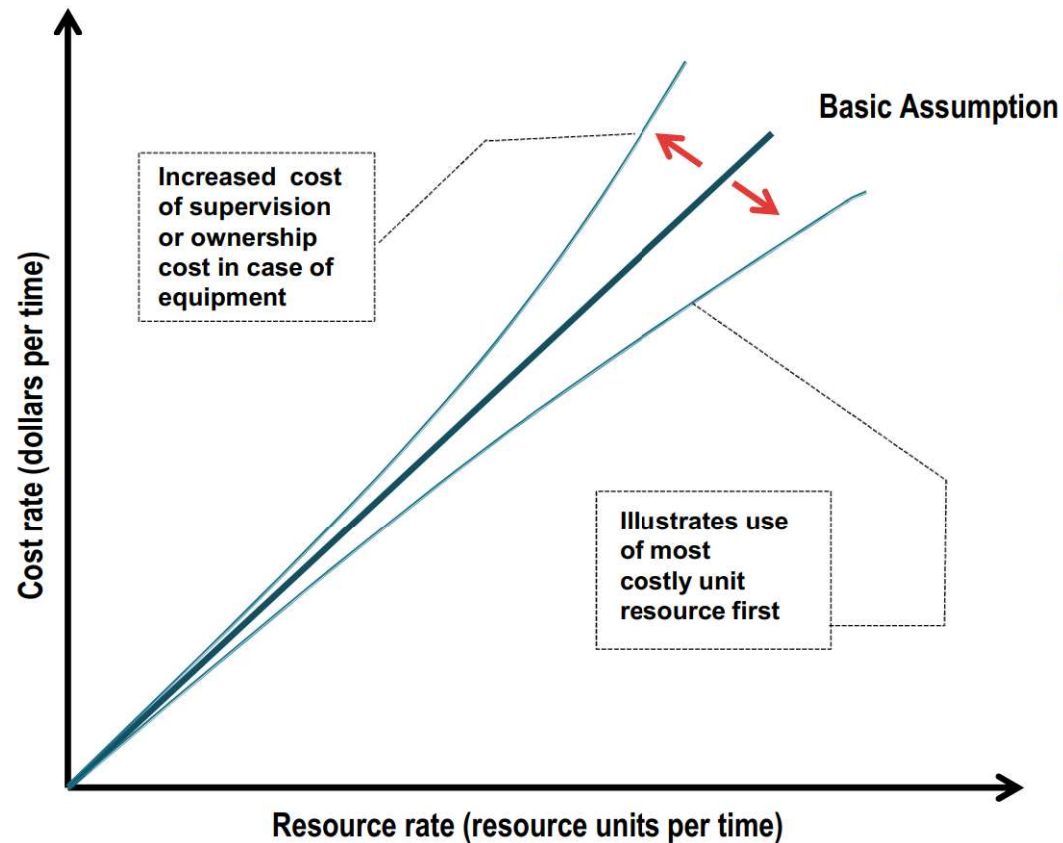
- Compress or “Crash” the schedule
- It consists of
 - Reducing the duration of tasks on the critical path
 - Increasing task parallelism
- This often comes at the expense of increasing project cost

STRATEGIES FOR REDUCING THE CRITICAL PATH BY REDUCING ACTIVITY DURATIONS

III

- Reallocate and/or add resources
 - Add Painters to paint activity
 - There is an upper limit on how much resources you can add. Too many resources will be ineffective and may actually result in longer performance period.
- Crashing the critical activities has some limitations
 - In some cases, an increase in number of resources results in a proportional activity duration reduction at no additional cost.
 - Many situations do not fit the linear model, however, and those that do are only effective over a limited range.
 - Compressing task duration may increase labor (overtime) and/or tooling cost.
- Example: a brick wall requiring 10 days for a single bricklayer to erect can probably be erected in 5 days by 2 bricklayers, but it is extremely unlikely that 10 bricklayers can accomplish the task in 1 day.

MODEL FOR ACTIVITY TIME-COST TRADEOFF

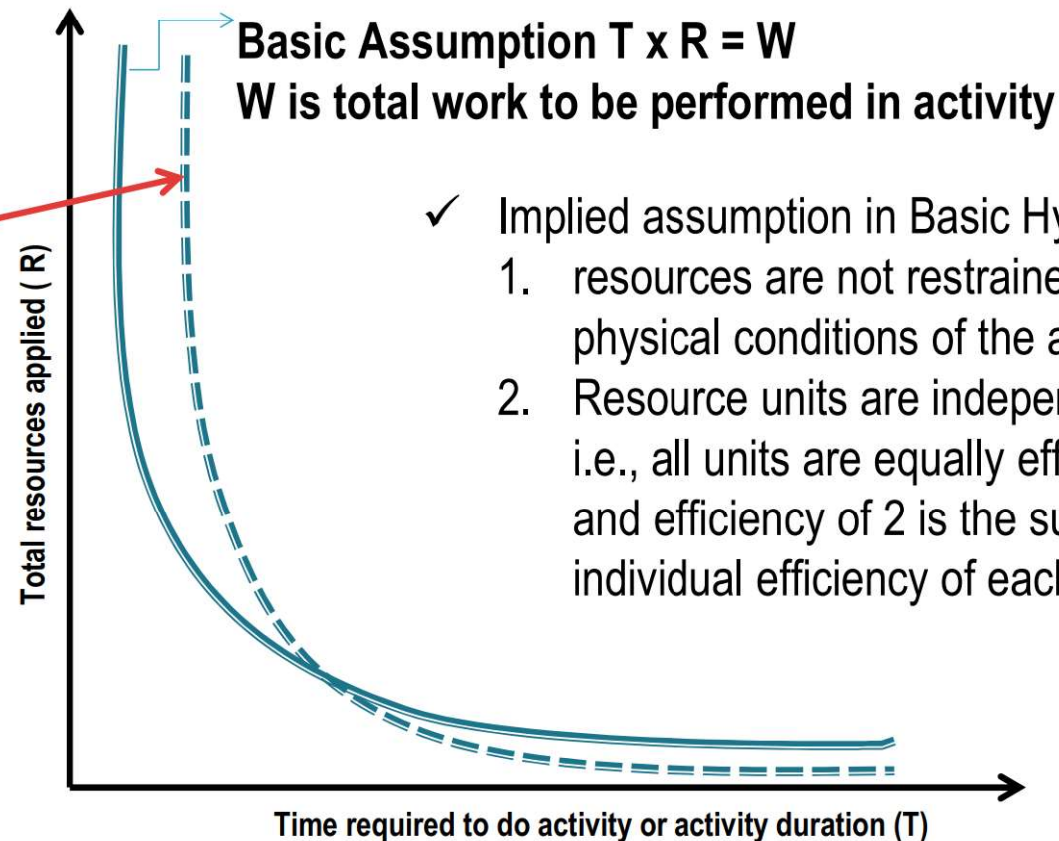


III

MODEL FOR ACTIVITY TIME-COST TRADEOFF

III

Deviations occur
due to inefficiencies

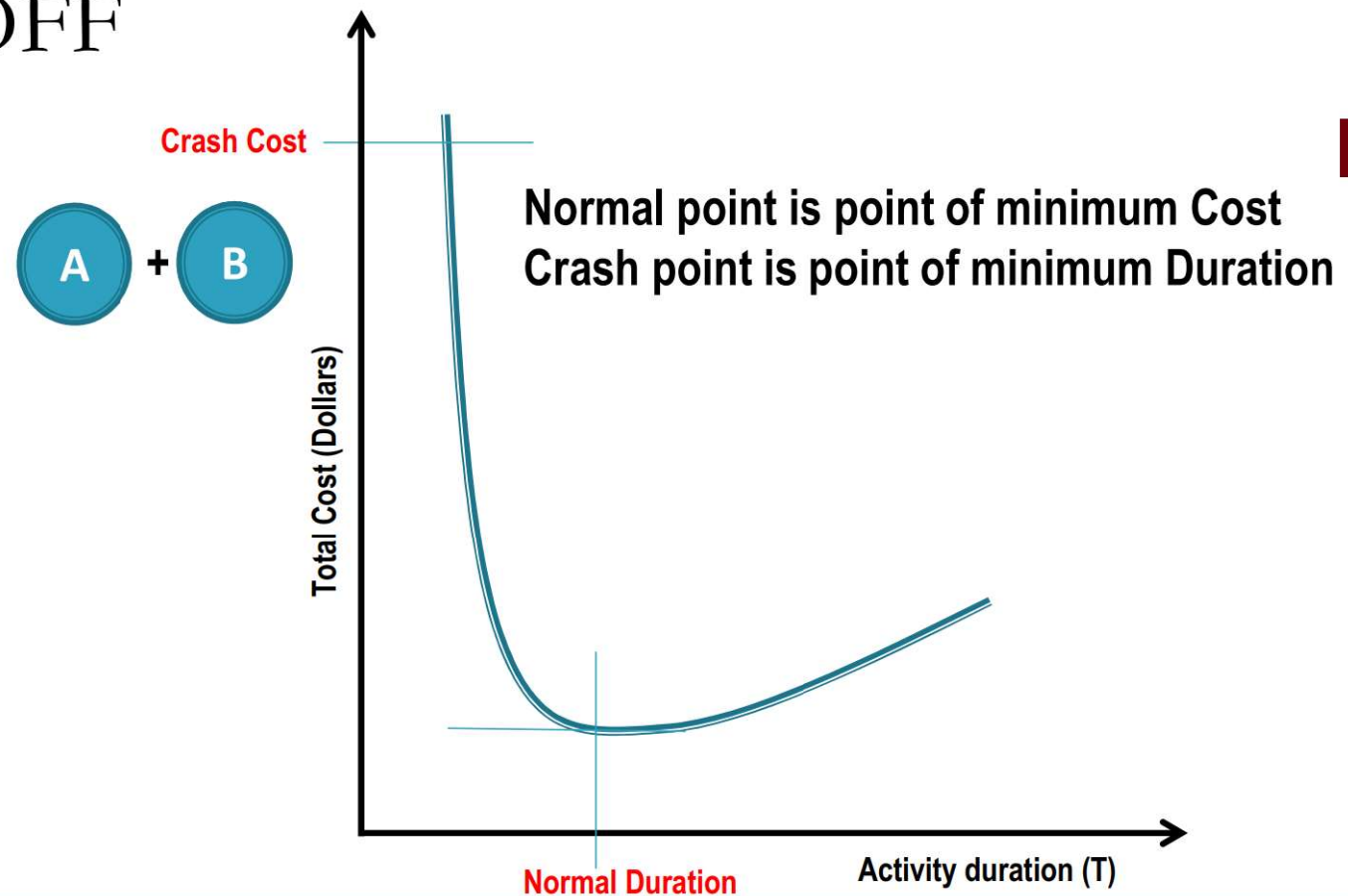


MODEL FOR ACTIVITY TIME-COST TRADEOFF

Laborers	Hourly rate	Remarks	Estimated activity time	Activity Total Cost
1	\$10		56 hours	\$560
2	\$20	Efficiency of 2 workers working will increase productivity by 25%	22.5	\$450
3	\$32	Third man has a longevity rate of \$12 per hour	18.7	\$600
4	\$44	Second \$12 per hour man added	17.5	\$770
5	\$59	Foreman added at \$14 per hour	16	\$944

III

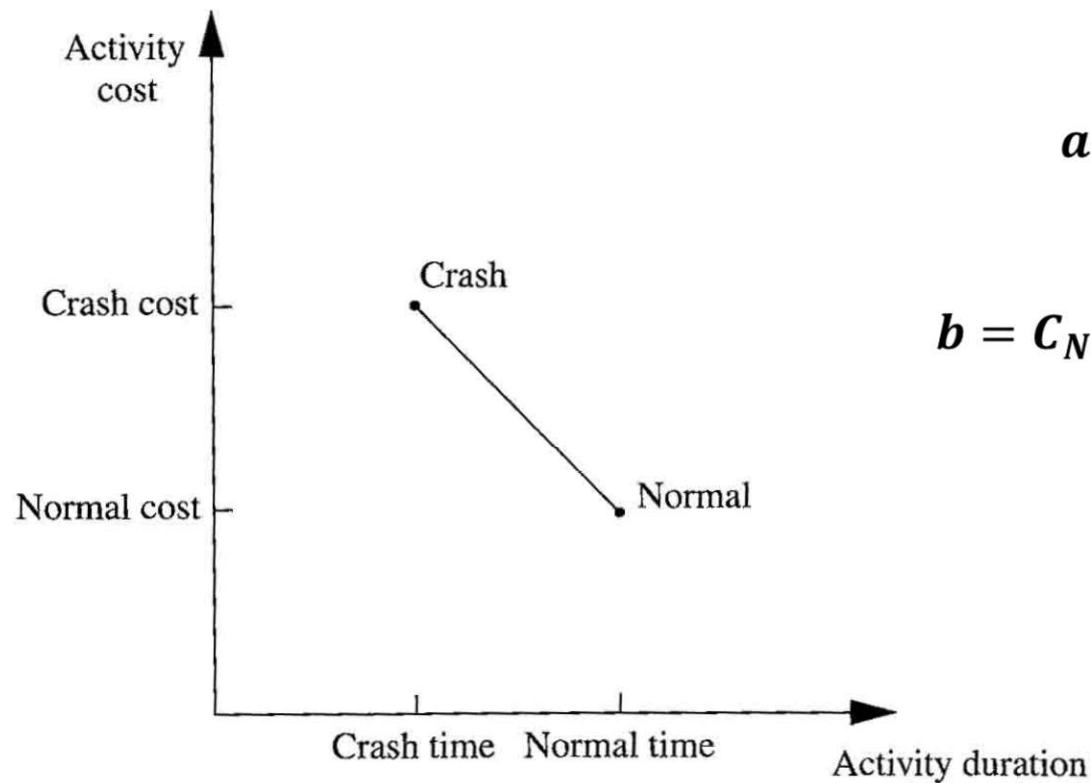
MODEL FOR ACTIVITY TIME-COST TRADEOFF



III

MODEL FOR ACTIVITY TIME-COST TRADEOFF

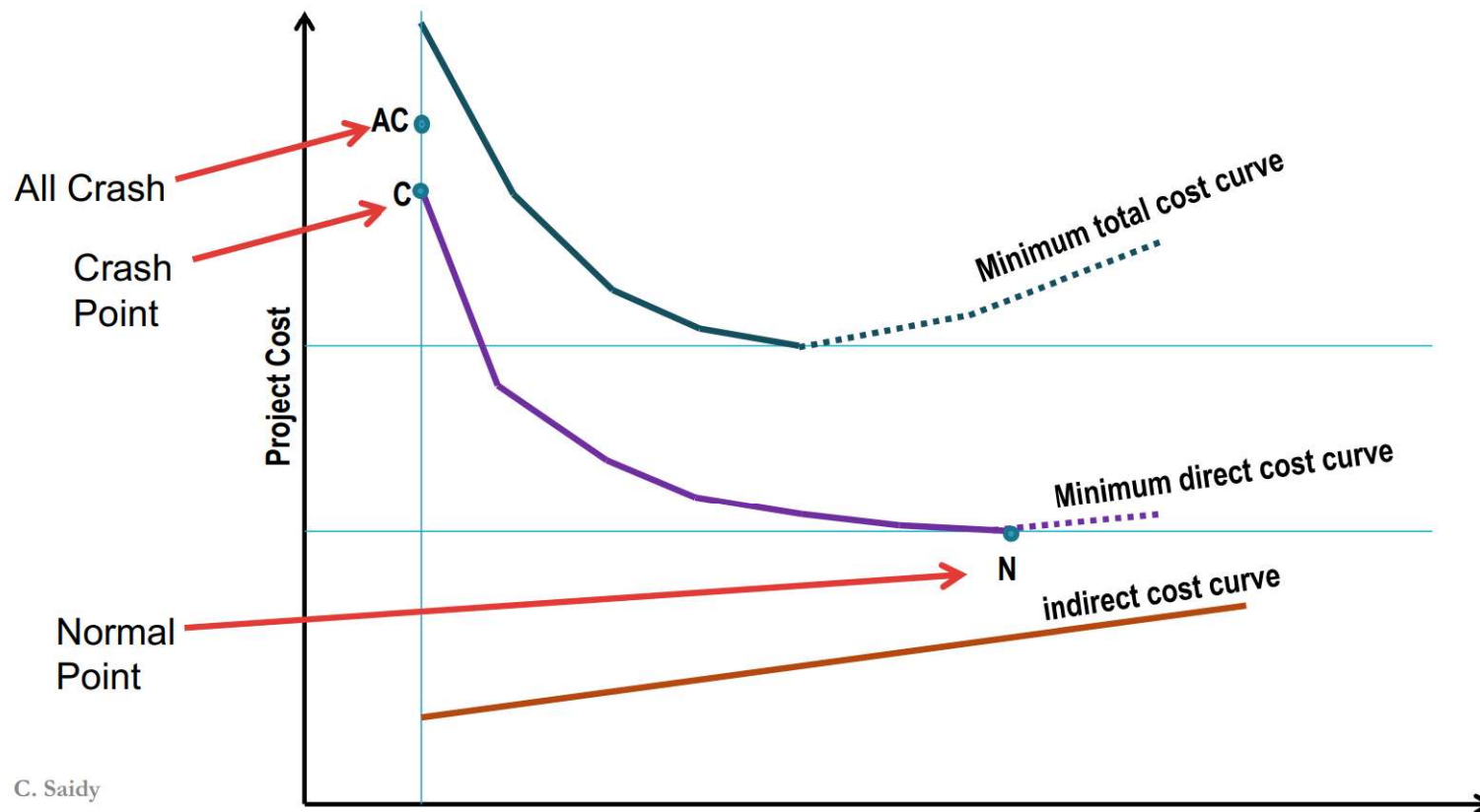
III



$$a = \frac{C_C - C_N}{T_C - T_N}$$

$$b = C_N - \left(\frac{C_C - C_N}{T_C - T_N} \right) T_N$$

PROJECT TIME-COST RELATIONSHIP



III

SPECIAL POINTS ON THE PROJECT TIME-COST CURVE

III

- The **Normal point** is the point of minimum cost on the direct time-cost curve for a project. All activities are performed with a normal level of resources applied.
- The **Crash point** is the point of minimum duration on the time-cost curve for a project.
- The **All-Crash** point is the point of minimum duration for a project where all activities are crashed by their maximum allowed limit.

METHODS FOR CRASHING THE PROJECT

III

- Heuristics
 - Select critical activity or activities that if crashed will shorten all existing critical paths
 - Activity(ies) can be crashed
 - If shortened, activity(ies) will bring the least increase in project cost
- Optimal Methods – Mathematical Programming

OPTIMAL FORMULATION FOR TIME-COST TRADEOFF

Let T_j be duration of activity j
 PD be desired/scheduled project duration

$$C_j = a_j T_j + b_j$$

Time-Cost Tradeoff Model

$$\text{Min } TC = \sum_{j=1}^n a_j T_j + b_j$$

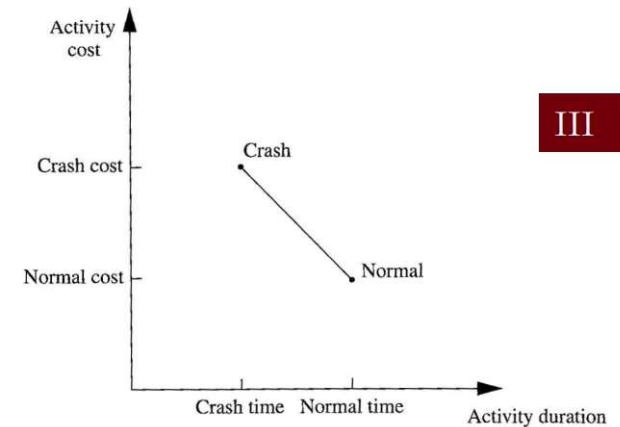
Such that

$$ES_0 = 0$$

$$ES_j \geq ES_i + T_i \quad ; \forall i \in P \text{ set of predecessors of } j$$

$$ES_{fin} + T_{fin} \leq PD$$

$$T_C \leq T_j \leq T_N$$



$$a = \frac{C_C - C_N}{T_C - T_N}$$

$$b = C_N - \left(\frac{C_C - C_N}{T_C - T_N} \right) T_N$$

COMPRESSING THE CRITICAL PATH

- In order to continually improve our process and reduce the time it takes to construct the small commercial building, we must constantly be looking for ways to compress the critical path
- To do this we must look for
 - ways to do tasks in parallel rather than in series,
 - apply technology to reduce the level of effort and time required, shift resources from non-critical to critical path tasks,
 - and/or increase number of resources allocated.

RESOURCE-CONSTRAINED SCHEDULING

III

- The number of units of a particular resource type is limited
- A particular resource is over-allocated
- The project must be finished as soon as possible, but without exceeding some specific level of resource usage.

RESOURCE-CONSTRAINED SCHEDULING

III

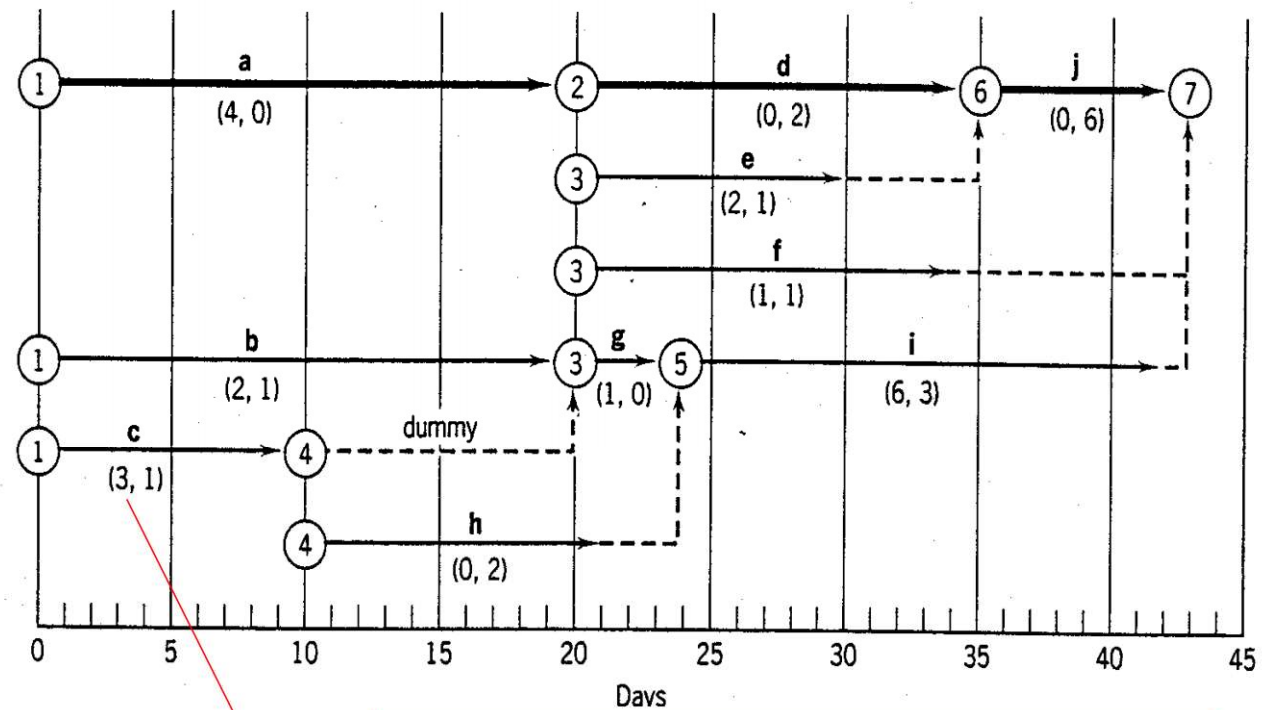
- A resource is **overloaded** when it is assigned too much work
- For example: an 80 person-hour task (effort), to be completed in 5 working days (duration)
 - if one full-time person was assigned, that person would be **overloaded** by 40 hours
 - if two full-time people were assigned, resources would be sufficiently allocated

RESOURCE-CONSTRAINED SCHEDULING

III

- If resources are **overloaded**, then one of the following must occur:
 - Add additional resources (procure more resources) to keep duration of task the same
 - Work overtime
 - Lower the effort per unit time by lengthening the task duration (in the previous example duration of task becomes 10 days)

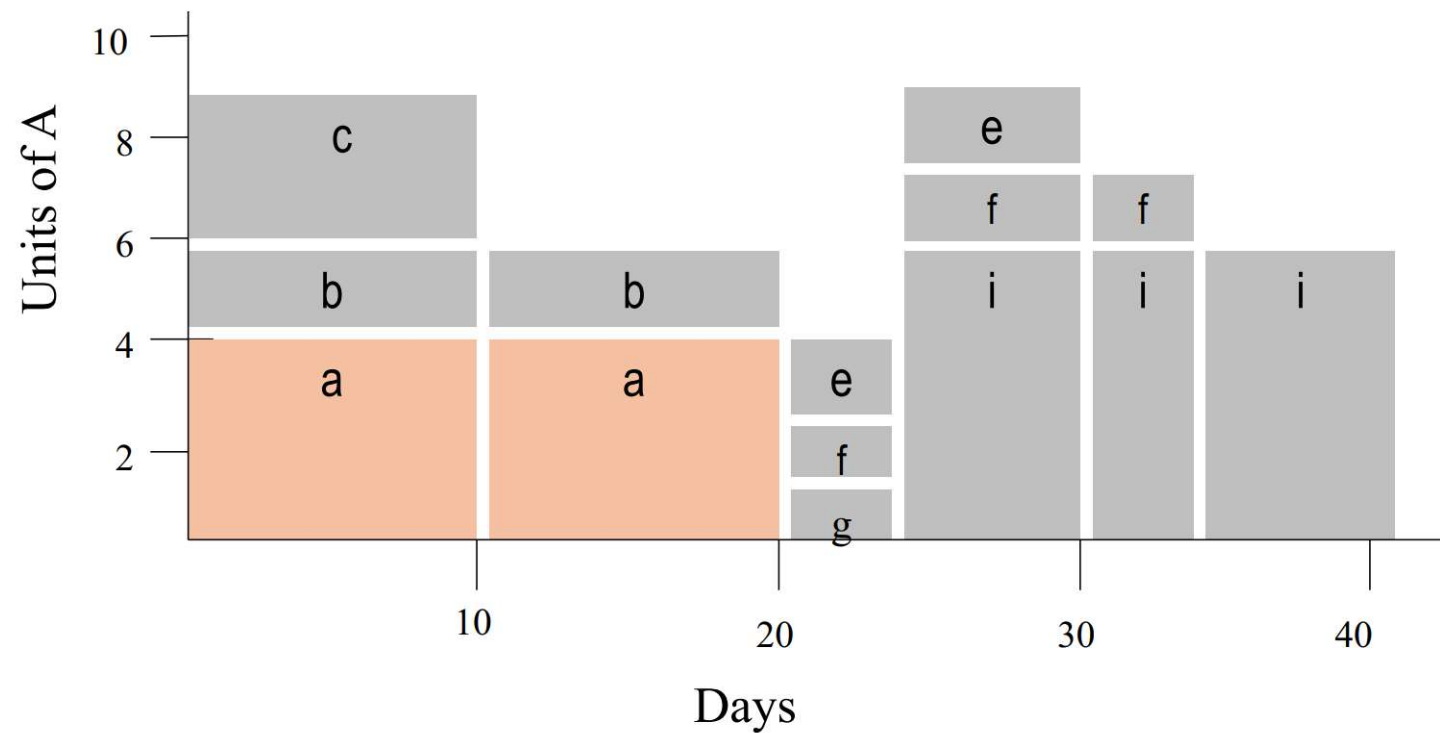
MODIFIED PERT DIAGRAM ANNOTATED WITH RESOURCE USAGE



III

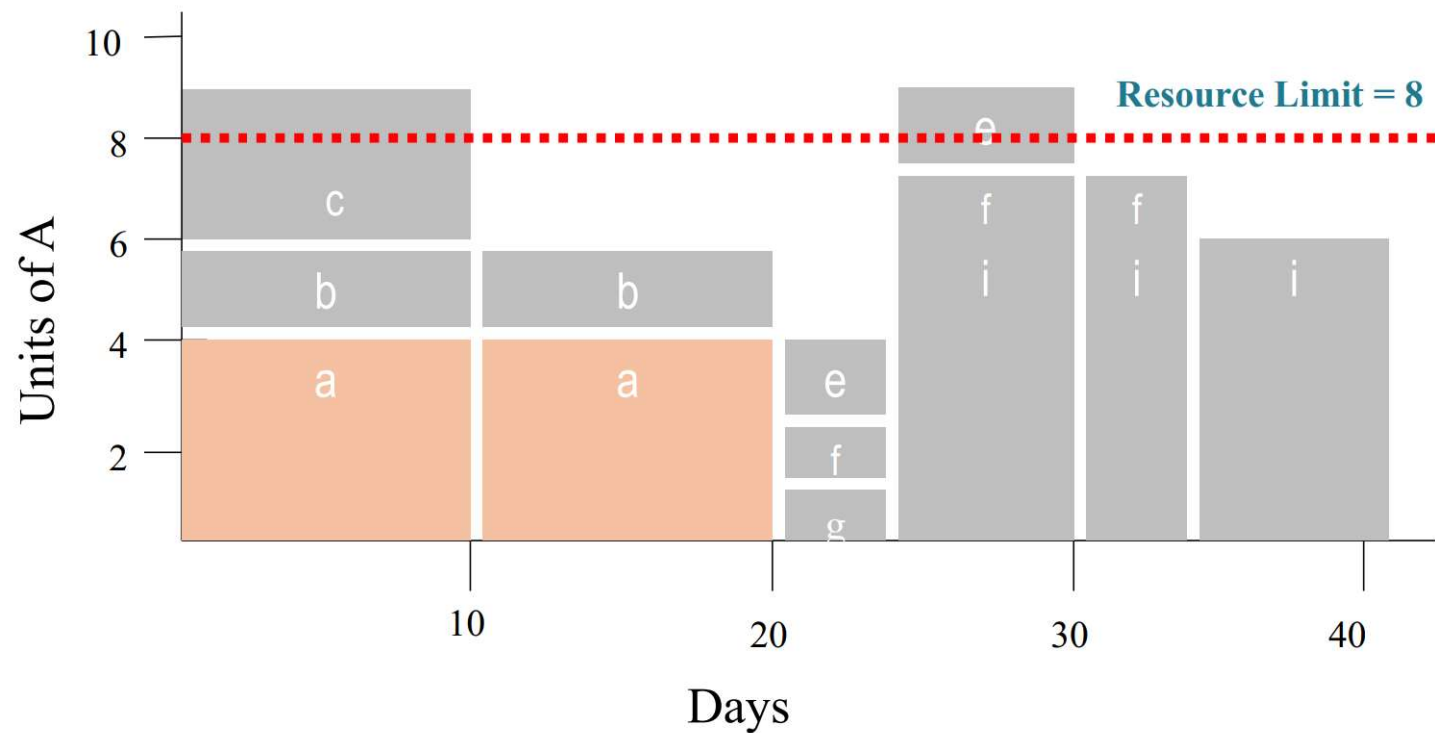
RESOURCE A HISTOGRAM FOR PERT DIAGRAM

III



RESOURCE A HISTOGRAM FOR PERT DIAGRAM

III



ACTIONS TO MAKE EXAMPLE SCHEDULE COMPLY WITH RESOURCE LIMIT ON RESOURCE A

III

- Delay C to start on Day 20, this will shift project completion date by 10 days.
- Perform C with 2 units of resource A instead of 3 units. This will increase its duration by 5 days and makes it 15 days.
- Delay E's start date by 10 days, this will delay project completion date by 1 day and lower the demand for resource A over the period between day 24 and day 30.

RESOURCE LEVELING

III

- Resource leveling aims at
 - minimizing fluctuations in resource use over time while holding the project duration the same
 - minimizing cost of hiring and firing in case of human resource; setup cost or rental costs in case of machine
- The result is a smoother distribution of resource usage

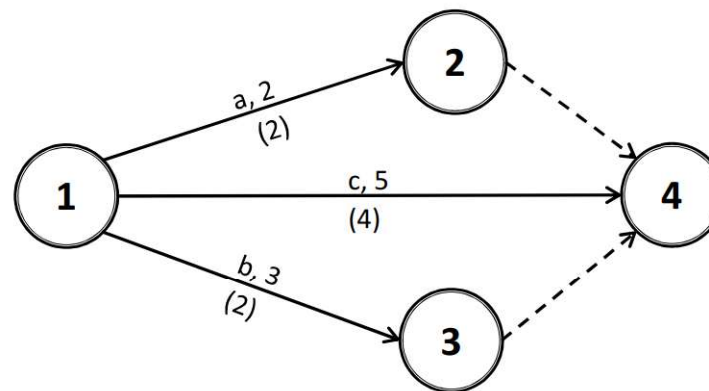
WHY DO RESOURCE LEVELING?

III

- Managerial implications
 - Reduce hand-on management
 - Plan for just-in-time material delivery
- Cost implications
 - Reduce cost of hiring and firing resources
 - Reduce equipment procurement cost
- Productivity implications
 - Learning curve effect

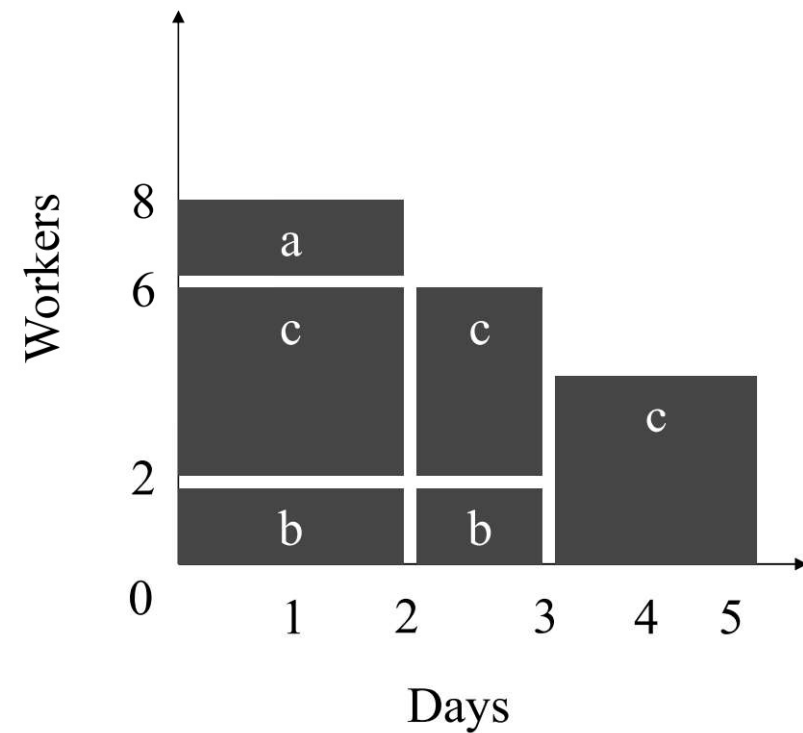
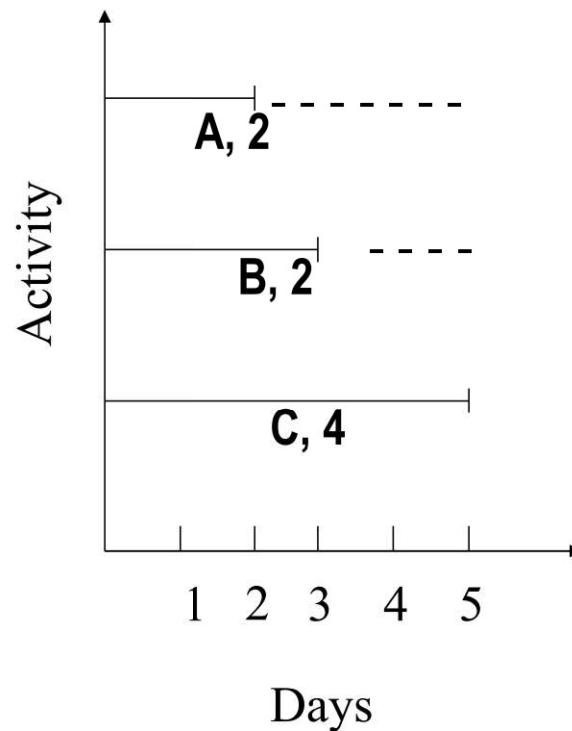
RESOURCE LEVELING – EXAMPLE

III

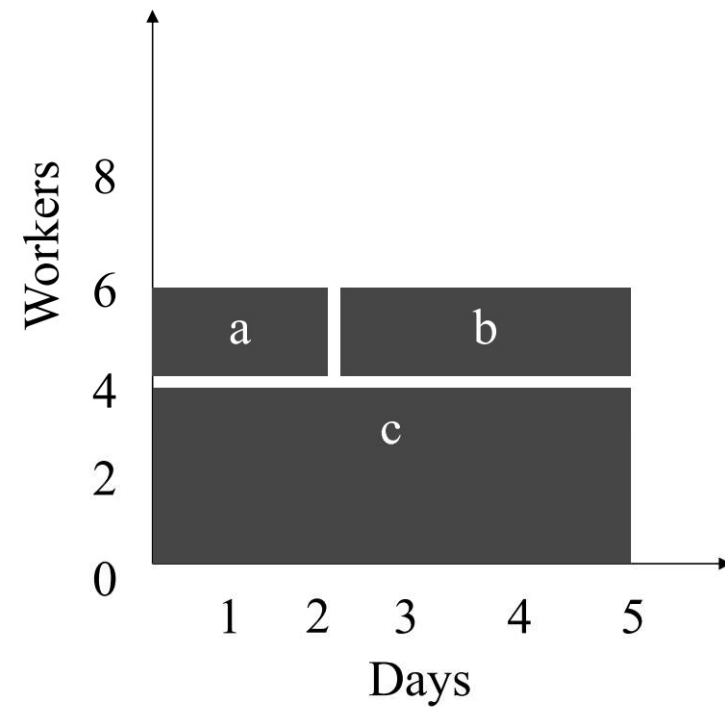
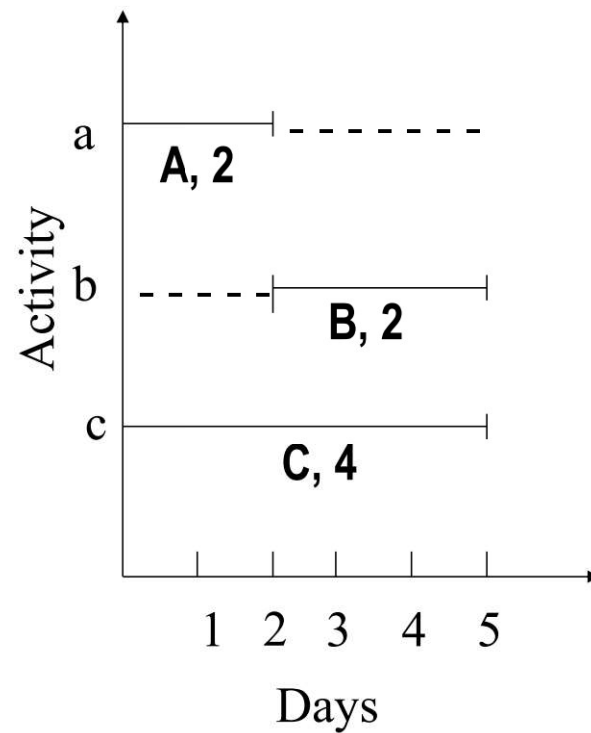


RESOURCE LEVELING – EXAMPLE BEFORE LEVELING

III



RESOURCE LEVELING – EXAMPLE AFTER LEVELING



III

Project Planning Wrap-up

Project Management – Project Scheduling

PROJECT PLANNING WRAP-UP

IV

- Review Past Experience/Templates
- Develop Work Breakdown Structure
- Establish Dependencies - Develop Logic Network
- Assign Resources
- Estimate Effort and Duration of Tasks
- Crash project duration if needed
- Level / Load Resources

WORK PLAN REVIEW

GAIN TEAM COMMITMENT & BUY-IN TO THE PLAN

IV

- During project and/or phases kick-off meetings
- Entire team involved
- Major focus:
 - Validate the work plan
 - Each member must review tasks assigned and validate effort and dependencies
 - Each member has the responsibility to ensure that work plan is correct and achievable
 - Gain team consensus
 - Emphasize importance of each member buying into the work plan
 - Surface unresolved issues
 - Now is time to correct any over estimations on effort, unrealistic dependencies, over constrained tasks, and the like.

WORK PLAN REVIEW

GAIN TEAM COMMITMENT & BUY-IN TO THE PLAN

- Generate action plans
 - Update work plan
 - Resolve open issues

IV

EFFECTIVE MEETINGS

- A meeting is an excellent tool for sharing information and building team consensus
- Preparation is critical to the success of a meeting
- Use the correct meeting format
 - Issues meeting (Team Meeting)
 - Working session (Sub-Team Meeting)
- Good meeting process should be observed

EFFECTIVE MEETINGS

- Adhere to effective communication precepts
 - Let everyone know objective of meeting
 - Understand what is to be accomplished
- Use the work plan to focus discussions
- Follow up with meeting minutes
 - Historical record of project issues and discussions
 - Decisions reached
 - Actions items

Thank you for your attention!

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